



Oxford Royale Academy
inspirational cultural and academic courses



Computer Science



Oxford Royale Academy

inspirational cultural and academic courses

16-18 Programme

Computer Science Programme

Student's Guide

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VERSION CONTROL FORM

Version 1.0. 2018.Feb.27 16-18 Computer Science Programme: Student's Guide.

It is Oxford Royale's responsibility to ensure that every staff member involved in the development and teaching of courses is familiar with the versions of this document.

The document is subject to regular revision, and maintained electronically. Electronic copies are version controlled.

CHANGE CONTROL DETAILS

SECTION CHANGED/ADDED (DATE)	DESCRIPTION

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Welcome to the Oxford Royale Academy Computer Science Programme!

Dear student,

This binder is designed to facilitate your learning during the Computer Science Programme at Oxford Royale Academy.

The format of the course is designed to promote self-directed learning and independence. It is divided into lectures, labs and project work. The lectures require you to take your own notes and to be proactive in asking questions. Most lectures contain a problem, which promotes the critical thinking skills required in computer science (and ensures that you pay attention!). You are encouraged to work together to solve the problems. The labs are an opportunity for you to get hands-on experience in programming. While you should work through the problems individually and at your own pace, it is expected that around half the time should be devoted to the exercises and half to the (open-ended) project work. The project work allows you to demonstrate what you have learnt, whether it be a solid understanding of the basics or a challenging extension. This course structure bridges the gap between school and university teaching techniques.

The assessment is divided equally into five areas. You will be assessed by means of a project which runs throughout the first ten days of the course. In Week 2, you will be exposed to some teamwork strategies for testing and debugging, which you may use if you wish. However, it is expected that the content of the project is entirely your own work. There will be a presentation lesson later in the course. Please be prepared to answer questions!

Try your best, but above all, have fun!

From all the staff at ORA

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General course information and timetable

Course aims and objectives

During this course, you will:

- Understand the scope of modern computer science, including topical issues
- Practise successful methods for composing, testing and documenting computer programs, applicable to various computer languages
- Demonstrate writing and compilation of short programs in HTML 5, CSS and Python 3
- Familiarise yourself with the purpose and composition of the aforementioned programming languages

Learning outcomes

Informatics

You should understand:

- An overview of topical issues in computer science
- Data representation on a computer, including numerical errors
- Effective management and visualisation of large data sets
- The suitability of various different programming languages to different tasks

Scientific programming

You should demonstrate:

- The ability to write correct Python programs to manipulate data and solve simple problems
- An understanding of how to document, test and debug both your own and other users' code
- The ability to translate a problem from another language into suitable pseudo-code

Web programming

You should produce:

- A summary of modern internet technologies
- Standards-compliant content written in HTML 5.0
- Standards-compliant formatting via CSS 3.0

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Transferable skills

You should acquire:

- Time management skills
- The ability to produce qualitative estimates
- An understanding of good presentation techniques
- Effective communication techniques in a variety of text types and media

In summary, you should be able to use computers effectively to analyse, present and share data with your peers. You should develop an appreciation for the utility and breadth of the field of computer science.

Assessment

The mark scheme for all Oxford Royale Academy courses is as follows:

Criterion	Weighting towards final grade	Marking scale
Programming tasks	30%	1-30
Presentation and report	30%	1-30
Contribution of ideas in class	20%	1-20
Progress/effort	10%	1-10
Attendance and punctuality	10%	1-10

The marks you receive will combine to make a total out of 100, determining your final grade:

Grade	A*	A	B	C	D	E	F (fail)
Total score	> 90	80-89	70-79	60-69	50-59	40-49	< 40

If you score below 40 marks, you are deemed to have failed the course and will not be eligible for graduation.

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Written work – marking criteria

Computer science (Max. 9 marks)	Structure and organisation (Max. 9 marks)	Content/research (Max. 12 marks)
7-9 marks High level of accuracy and all work completed. Mistakes occur only when the student is attempting the more ambitious problems.	7-9 marks Ideas are organised clearly throughout and each section of the argument is clearly signposted for the reader.	9-12 marks The topic is explored thoroughly and comprehensively. The student has mastered the subject area of the assignment.
4-6 marks Work not fully completed and/or not fully accurate. Demonstration of subject ability is solid.	4-6 marks The piece of work has a clear introduction and conclusion and the student is able to link their ideas logically.	5-8 marks The topic is explored in detail and a generally solid understanding is shown.
1-3 marks Insufficient effort demonstrated with work incomplete or inaccurate.	1-3 marks The writing lacks a clear structure and it is not always clear how the points the student is making are linked.	1-4 marks Some key subject areas are not covered or there is evidence of lack of understanding.

Interactive assignments/presentations – marking criteria

Computer science (Max. 9 marks)	Organisation and delivery (Max. 9 marks)	Content/research (Max. 12 marks)
7-9 marks High level of accuracy and all work completed. Mistakes occur only when the student is attempting the more ambitious problems.	7-9 marks Ideas are organised clearly throughout and delivery is confident, fluent and engaging.	9-12 marks The topic is explored thoroughly and comprehensively. The student has mastered the subject area of the assignment.
4-6 marks Work not fully completed and/or not fully accurate. Demonstration of subject ability is solid.	4-6 marks An effort has clearly been made structure the presentation to maximise the audience's engagement. However, there may be some hesitation in the delivery.	5-8 marks The topic is explored in detail and the student demonstrates a solid understanding of the most important aspects of the topic.
1-3 marks Insufficient effort demonstrated with work incomplete or inaccurate.	1-3 marks Organisation is weak and there are frequent pauses, hesitations and blunders in the delivery.	1-4 marks Some key subject areas are not covered or there is evidence of lack of understanding.

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Assessed work

You will complete a major project which consists of 2 parts: lab work and a presentation. The lab work includes programming in Python and web development.

Major project

Progress in labs

These marks give you the opportunity to demonstrate your practical skills, rather than your acquired knowledge. Both lab tasks will receive a total mark out of 15 – these will then be combined to produce a final mark of 30 towards your overall final grade.

The major project tests your hands-on computer skills. It will take up the majority of the time spent in the computer labs. The aim is to solve a real-world problem using the skills covered in the course. You will write a short webpage on your work and give a brief presentation on an aspect of the problem your project is solving.

Format

The project is in four stages:

1. Problem-solving using Python
2. Composition of a webpage report via markup in HTML and styling using CSS
3. Report
4. Presentation during class- Wednesday Week 2

Topic

You will be required to propose your own project; however, it has to be approved by the teacher. You may do one or more projects depending on how comfortable you are with the material. However, you are encouraged to improve the sophistication of a project in focus.

Timetable

The suggested deadlines are as follows:

Webpage	Friday afternoon of Week 1 (Lab III)
Presentation	Wednesday morning of Week 2 (Lessons 19 & 20)
Report	Wednesday afternoon of Week 2 (Lab V)
Python program	Wednesday afternoon of Week 2 (Lab V)

If you fall behind, do not panic: ask for help from your teacher. You are responsible for your own time management. The teacher will not chase you to ensure that your project is progressing well.

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Webpage

You are required to create a website selling a futuristic product which does not yet exist but would be a great seller. The website should describe this product, how it works and why it would be such a bestseller. You should also include images of the said product, either hand-drawn or created with the help of CAD. There are two stages to this section of the project: formatting and markup. It is possible to do much of the work before finishing the lessons on web programming. For example, it is possible to decide on a page layout, font and colour scheme on paper independent of the methods to enforce them (for more details, see Lab 1).

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Python program

You are required to develop a Python program that solves a particular problem. You must propose your own project, but it has to be approved by the teacher (for more details, see Lab 3).

Examples of projects you can work on are:

- Currency converter
- Programming a smart fridge

Presentation

The presentation is to be given on Wednesday of Week 2. Your aim is to teach your peers something about your Python software development process.

- What problem were you solving?
- Did you find a clever method to simplify the problem?
- Did you use a particularly efficient algorithm used?
- How would you adjust the process for a more realistic problem?
- What would you have done differently if you had had more time/computing power?

Each talk should be 10 -12 minutes long. You may use PowerPoint.

Report

You are required to write an 750-1000 word report summarising the problem solved by your Python program. The style of the report should be that of an academic journal article with an abstract, introduction, conclusion and references. It is suggested that you write as you go, making comments to document your code, or leave perhaps half an hour at the end of each lab session to make notes on what you and your partner completed that afternoon.

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Plagiarism guidelines

Anti-plagiarism rules at ORA

Plagiarism is **never** tolerated at Oxford Royale Academy. If any piece of work submitted by a student is found to be plagiarised, serious disciplinary measures will be taken.

Strike one: If a piece of work submitted is found to contain plagiarism, your teacher will report this to your Programme Director as a disciplinary matter.

You will be given the opportunity to re-submit your work **without** plagiarism, but your final mark for this work on your report will be reduced by 50%. For example, if your second piece of work receives 24 marks, this will be reduced to 12 marks on your final report.

Strike two: If a second piece of work submitted is found to contain plagiarism **or** no second piece is submitted, you will fail the course and may be barred from graduation.

What is plagiarism?

Available from <http://www.plagiarism.org/plagiarism-101/what-is-plagiarism> (accessed: 24th May 2016)

Many people think of plagiarism as copying another's work or borrowing someone else's original ideas. However, terms like "copying" and "borrowing" can disguise the seriousness of the offence.

According to the *Merriam-Webster Online Dictionary*, to "plagiarise" means:

- To steal and pass off (the ideas or words of another) as one's own
- To use (another's production) without crediting the source
- To commit literary theft
- To present as new and original an idea or product derived from an existing source

In other words, plagiarism is an act of fraud. It involves both stealing someone else's work and lying about it afterwards. At university, it can result in expulsion and a permanent record.

Never attempt it, not even as a shortcut. It is very easy to spot, especially if you are writing in your second language. Your Oxford Royale Academy teachers, and in the future your university tutors, will be able to tell whether your written work matches your understanding of the subject. At most universities today, all written submissions are also scanned through plagiarism software that can immediately detect any copied work.

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All of the following are considered plagiarism:

- Turning in someone else's work as your own
- Copying words or ideas from someone else without giving credit
- Failing to put a quotation in quotation marks
- Giving incorrect information about the source of a quotation
- Changing words but copying the sentence structure of a source without giving credit
- Copying so many words or ideas from a source that it makes up the majority of your work, whether you give credit or not

Further information on types of plagiarism, and examples of what to avoid, can be found at <http://en.writecheck.com/plagiarism-guide>.

Avoiding plagiarism

The rule of thumb in avoiding plagiarism is this: always ensure that you acknowledge whose material you are using. Here are some strategies for avoiding plagiarism:

- Clearly reference by using quotations for everything that comes directly from the text, especially when taking notes
- Paraphrase but make sure that you are not just rearranging or replacing a few words. Instead, read over what you want to paraphrase carefully, then cover up or close the text so you cannot see any of it and so will not be tempted to use the text as a guide
- Check your paraphrase against the original text to make sure you have not accidentally used the same phrases or words, and that all the information is accurate

If you are worried about plagiarism, ask your teacher, and always remember to acknowledge your sources.

How to cite secondary material in your work

After you have mentioned an author's idea or quoted directly from their work, you should cite the author's surname(s) and the year their book, article, etc., was printed, e.g.:

The Brayford Pool swan population at Lincoln was always in danger whenever grants were delayed (Duck, 2004).

At the end of any essay, you should include a Bibliography section that lists every item you have referred to in your essay. If you want to cite a webpage, you should include the date that you accessed the website, e.g.:

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Cotton, A. (2001), 'I love Lincoln' [online], available from: http://www.hero.ac.uk/uk/studying/archives/2001/i_love_lincoln1272.cfm (accessed: 26th October 2011)

Classroom conduct

You must be respectful at all times towards your teacher, ORA staff and other students. Please also remember to treat college staff and buildings with respect. You should always pay attention to safety announcements and fire drills. In the unlikely event of an emergency, remain calm and listen for instructions from your teacher and/or ORA staff.

Please be punctual for each and every class. Lateness is rude, both to your teacher and your fellow students. Your punctuality will form part of your overall assessment.

If at any time you feel unwell or there is something wrong, please tell your teacher immediately. Staff are here to help you.

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Course timetable

Computer Science 16-18										
	Week 1					Week 2				
	Monday	Tuesday	Wednesday	Thursday	Friday	Monday	Tuesday	Wednesday	Thursday	Friday
0900-1030	Lesson 1 Welcome and setup	Lesson 4 Tables and multimedia	Lesson 6 HTML forms and user input	Lesson 9 Welcome to Python	Lesson 11 Control statements	Lesson 14 Methods and functions	Lesson 17 Strings and REGEX	Lesson 19 Presentations	Lesson 22 Debugging	Lesson 25 Python challenge
1100-1230	Lesson 2 Introduction to HTML5	Lesson 5 CSS web styling	Lesson 7 Web scripting: JavaScript	Lesson 10 Variables and operators	Lesson 12 Iterations and conditions	Lesson 15 Data structures	Lesson 18 File operations	Lesson 20 Presentations	Lesson 23 Software testing	Lesson 26 Python challenge
1345-1530	Lesson 3 Lab 1: Project		Lesson 8 Lab 2: Project		Lesson 13 Lab 3: Project	Lesson 16 Lab 4: Project		Lesson 21 Lab 5: Project	Lesson 24 Evolution and maintenance	

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Lesson plans

Lesson 1: Welcome and setup

Learning objectives:

- To understand the structure of the course and learning aims for the next two weeks
- To get all the necessary installations done on the computers in preparation for the rest of the course
- To understand what is expected of the web programming assignment and presentation due on Friday and next Wednesday respectively
- To create your first webpage

Lesson outline:

Icebreaker

- With the person next to you, introduce yourself and talk about why you wanted to do the Computer Science course at Oxford Royale Academy
- Then, introduce yourselves to the class and explain:
 1. Why you both picked the Computer Science course
 2. What you both are most excited to learn from the course
 3. What your aims are for the next two weeks

Classroom rules

- In groups, come up with three classroom rules and then discuss them as a class
- This is a student-led learning environment and the rules are to enable all of you to learn

Web development assignment

- Your teacher will explain what you have to do for the web development assignment due on Friday

Get the tools

- Get a computer and install Brackets IDE: <http://brackets.io/>

First website demonstration

Your teacher will:

- Launch Brackets and demonstrate the HTML skeleton
- Add a paragraph to the body to display “Hello ORA” then demonstrate saving the file as an HTML file

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Exercise 1: My first website

- Using the additional support materials below and following your teacher's example, your task is to create your first HTML page in 10 minutes
- Make use of basic HTML skeleton (syntax) to write a short paragraph about yourself
- Run your codes and preview your webpage in a browser of your choice

Extra

- What do you need to create webpages?
- Which two tags are required for every webpage?
- What is the purpose of the <head> tag?
- What software do you need to view webpages that you create? List 5 types of software

Additional support materials

My first website

This is a HTML skeleton which you can use to create every webpage.

```
<!DOCTYPE html>
<html>

  <head>
    <title></title>
  </head>

  <body>

  </body>

</html>
```

Create your first website and display “Hello Wold” as shown.

```
<!DOCTYPE html>
<html>

  <head>
    <title>My first website</title>
  </head>

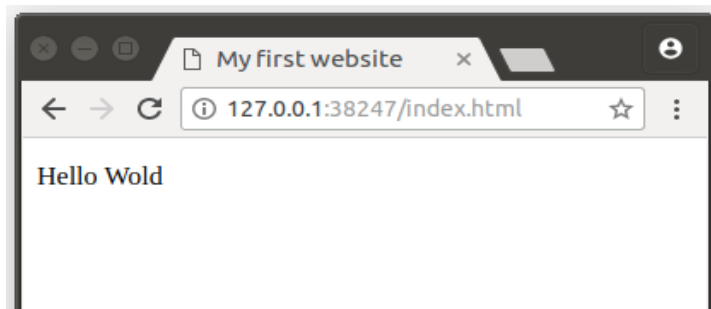
  <body>
    <p>Hello Wold</p>
  </body>

</html>
```

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When you run your website, it should display as shown below.



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Lesson 2: Introduction to HTML5

Learning objectives:

- To learn how to add elements to your html page
- To understand the different tags: <!DOCTYPE html>, <html>, <head>, <title>, <body> and <p>
- To complete an exercise on adding paragraphs, headings, hyperlinks, ordered and unordered lists

Lesson outline:

The HTML skeleton explained

Your teacher will:

- Describe the different tags in the HTML skeleton available in the workbook
- Explain how user defined tags are also allowed and may be necessary in a personal website

Adding paragraphs and headings

Your teacher will:

- Demonstrate how to add paragraphs
- Demonstrate the different HTML headings from H1 to H7 and explain their importance

Exercise 2

You have 7 minutes to:

- Create one paragraph that describes what you would like to be in the future
- Create two or more different headers in your story
- Run your webpage in a browser to view the output

Extra

- Display your headers in green and red
- Display your paragraph in “Courier” font

Adding hyperlinks

Your teacher will:

- Create a second HTML page
- Demonstrate how to add a hyperlink to navigate between pages
- Demonstrate how to add a hyperlink that references something online

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Exercise 3

You have 10 minutes to:

- Create a second decoy page containing any message of your choice
- Then, create a hyperlink on each page to navigate between pages
- Finally, create a hyperlink linking to a page online, for example, a Facebook or Wikipedia page

Extra

- Create some links to various search engines
- Create a link at the bottom of your webpage that jumps to the top of the page when clicked

Ordered and unordered lists

Your teacher will:

- Demonstrate how to add unordered lists
- Show you how to add an ordered list
- Highlight the difference between the two

Exercise 4

You have 5 minutes to:

- Create two lists of your choice (e.g., a list of food items or names). One should be an ordered list and the other an unordered list. Add both lists to your sample webpage

Extra

- Display 10 names of your classmates, on separate lines, in alphabetical order and in different colours

Additional support materials

Introduction to HTML5

Headings and paragraphs

The code below shows how to insert headings and paragraphs on your website.

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```
<!DOCTYPE html>
<html>
  <head>
    <title>This is a title</title>
  </head>

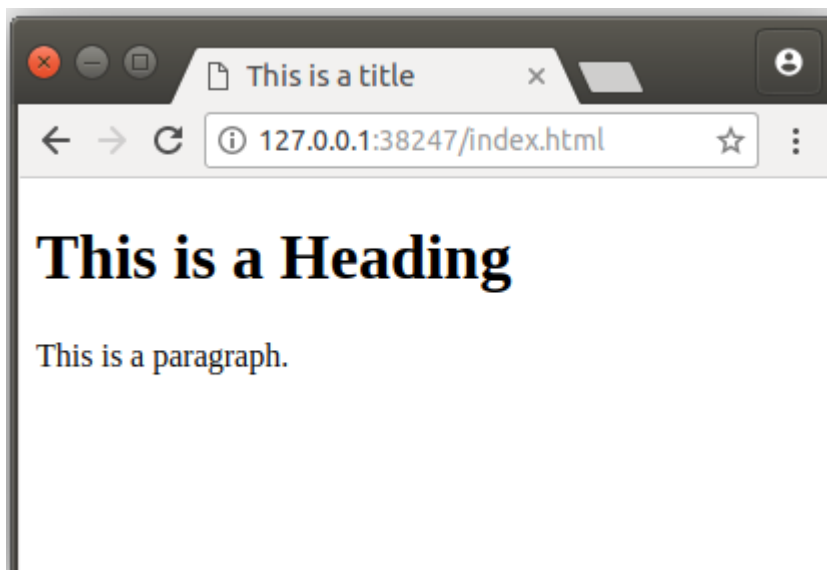
  <body>

    <h1>This is a Heading</h1>
    <p>This is a paragraph.</p>

  </body>

</html>
```

Your website should display as shown below.



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Hyperlinks

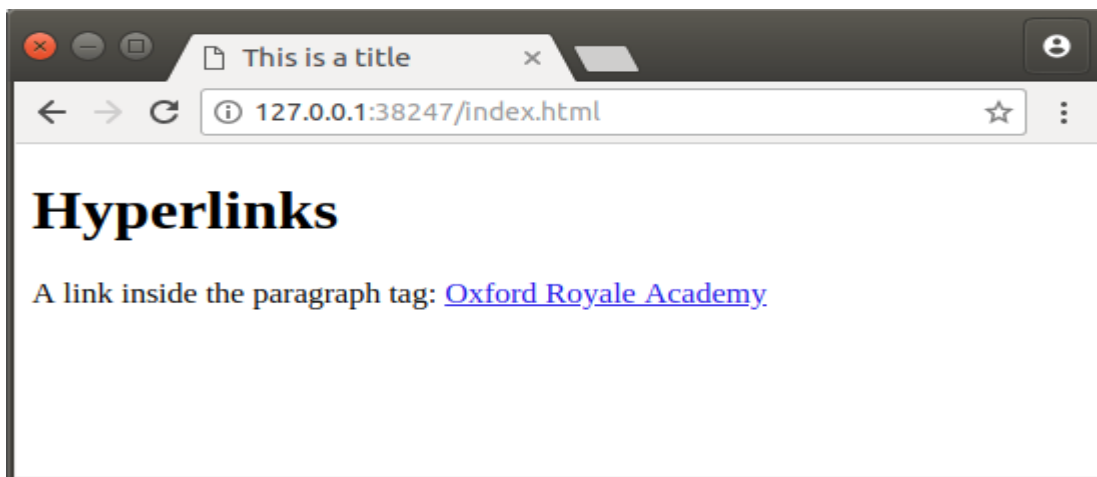
The code below shows how to create links on your website.

```
<!DOCTYPE html>
<html>
  <head>
    <title>This is a title</title>
  </head>
  <body>

    <h1>Hyperlinks</h1>
    <p>
      A link inside the paragraph tag:
      <a href="https://www.oxford-royale.co.uk/">Oxford Royale Academy</a>
    </p>

  </body>
</html>
```

When you run your website, it will look as shown below.



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Lists

The code below shows how to insert ordered and unordered lists on your website.

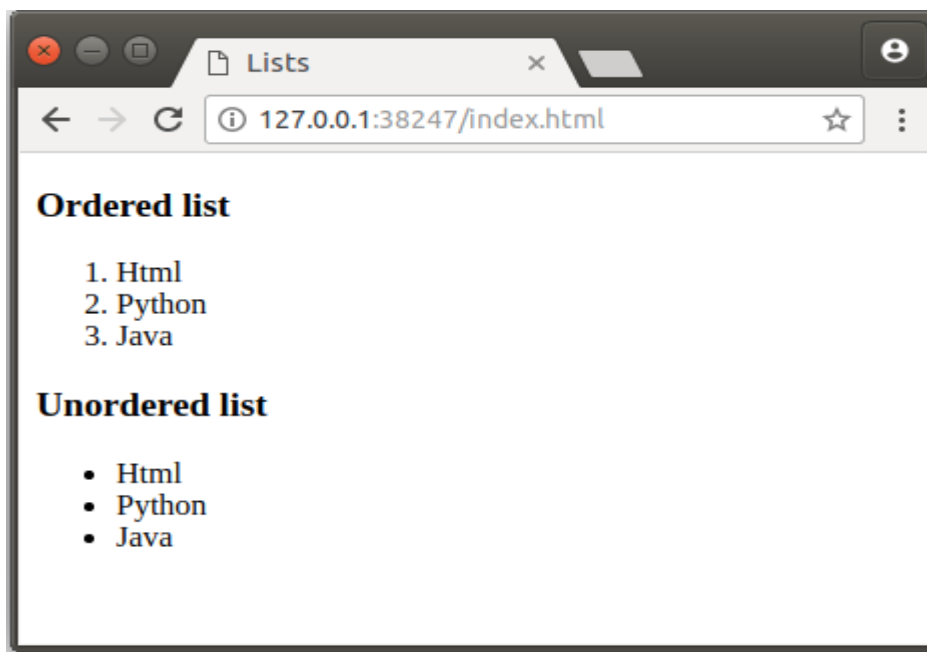
```
<!DOCTYPE html>
<html>
  <head>
    <title>List</title>
  </head>
  <body>

    <h3>Ordered list</h3>
    <ol>
      <li>Html</li>
      <li>Python</li>
      <li>Java</li>
    </ol>

    <h3>Unordered list</h3>
    <ul>
      <li>Html</li>
      <li>Python</li>
      <li>Java</li>
    </ul>

  </body>
</html>
```

When you run your website, it will look as shown below.



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Lesson 3: Lab 1 project

Learning objectives:

- To work on your web development project

Website development project outline

You are required to create a website selling a futuristic product which does not yet exist but which would be a great seller. The website should describe this product, how it works and why it would be such a bestseller. You should also include images of the said product, either hand-drawn or created with the help of CAD. There are two stages to this section of the project: formatting and markup. It is possible to do much of the work before finishing the lessons on web programming. For example, it is possible to decide on a page layout, font and colour scheme on paper independent of the methods to enforce them.

The webpage should incorporate the following features:

- At least one figure
- Formatted text
- At least one table or nested list
- Captions for all tables and figures

Your style sheet should obey basic principles of design:

- High contrast between the text and the background
- Some use of colour to add interest, e.g. on section headings
- Styled `<em>` and `<strong>` tags, not `<i>` and `<b>` tags

Make some notes about your code if you need to and feel free to talk to your teacher if you have any questions.

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Lesson 4: Tables and multimedia

Learning objectives:

- To learn how to create tables and use them to organise your webpage
- To learn how to add images, videos and animations to your website

Lesson outline:

Creating a table

Your teacher will:

- Demonstrate how to create a table
- Show you how to add rows to your table and data to populate them

Exercise 5

You have 10 minutes to:

- Create one table with at least two rows
- In each row, you must add table data tags and fill them with content

Extra

- You can format the table by adding borders or changing the column width
- Change the font style of your column headings

For example:

City	Country	Continent
Paris	France	Europe
Tokyo	Japan	Asia

Adding images

Your teacher will:

- Demonstrate how to add local images located on the computer
- Demonstrate how to add an image located on the internet by means of its URL

Exercise 6

You have 10 minutes to:

- Add two local images (images stored on your computer) to your webpage
- Add one image hosted online

Each image you have added to your website should have a title.

Extra

- Display a fourth image that when clicked, will link to your second decoy webpage or a search

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engine of your choice. Add a description below the image informing the user that the image is link

- Double the width and height of one of your images

Adding videos

Your teacher will:

- Demonstrate the different ways you can add videos to your website
- Demonstrate how to get a video from YouTube using the embed function

Exercise 7

You have 10 minutes to:

- Add one video stored on your computer to your website
- Add one YouTube video

Extra

- Add the “autoplay” attribute to one of the videos on your webpage

Adding animations

Your teacher will:

- Demonstrate how to add a GIF using the image tag

Exercise 8

You have 5 minutes to:

- Add one or more animated GIFs to your webpage
- Align your GIFs and add titles in text
- Run your codes and view your website in your browser

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Additional support materials

Tables and multimedia

Tables

The code below shows how to insert tables.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Tables</title>
  </head>
  <body>

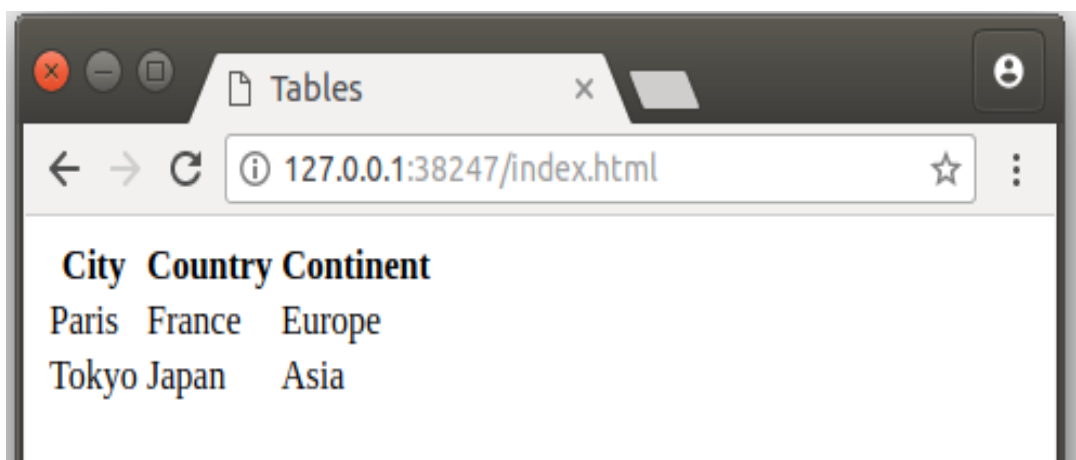
    <table>
      <tr>
        <th>City</th>
        <th>Country</th>
        <th>Continent</th>
      </tr>

      <tr>
        <td>Paris</td>
        <td>France</td>
        <td>Europe</td>
      </tr>

      <tr>
        <td>Tokyo</td>
        <td>Japan</td>
        <td>Asia</td>
      </tr>
    </table>

  </body>
</html>
```

When you run your website, it will look as shown below.



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Multimedia – Images, videos and animations

The code below shows how to insert images, videos and animations on your website.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Multimedia</title>
  </head>
  <body>

    <h3>Image</h3>
    

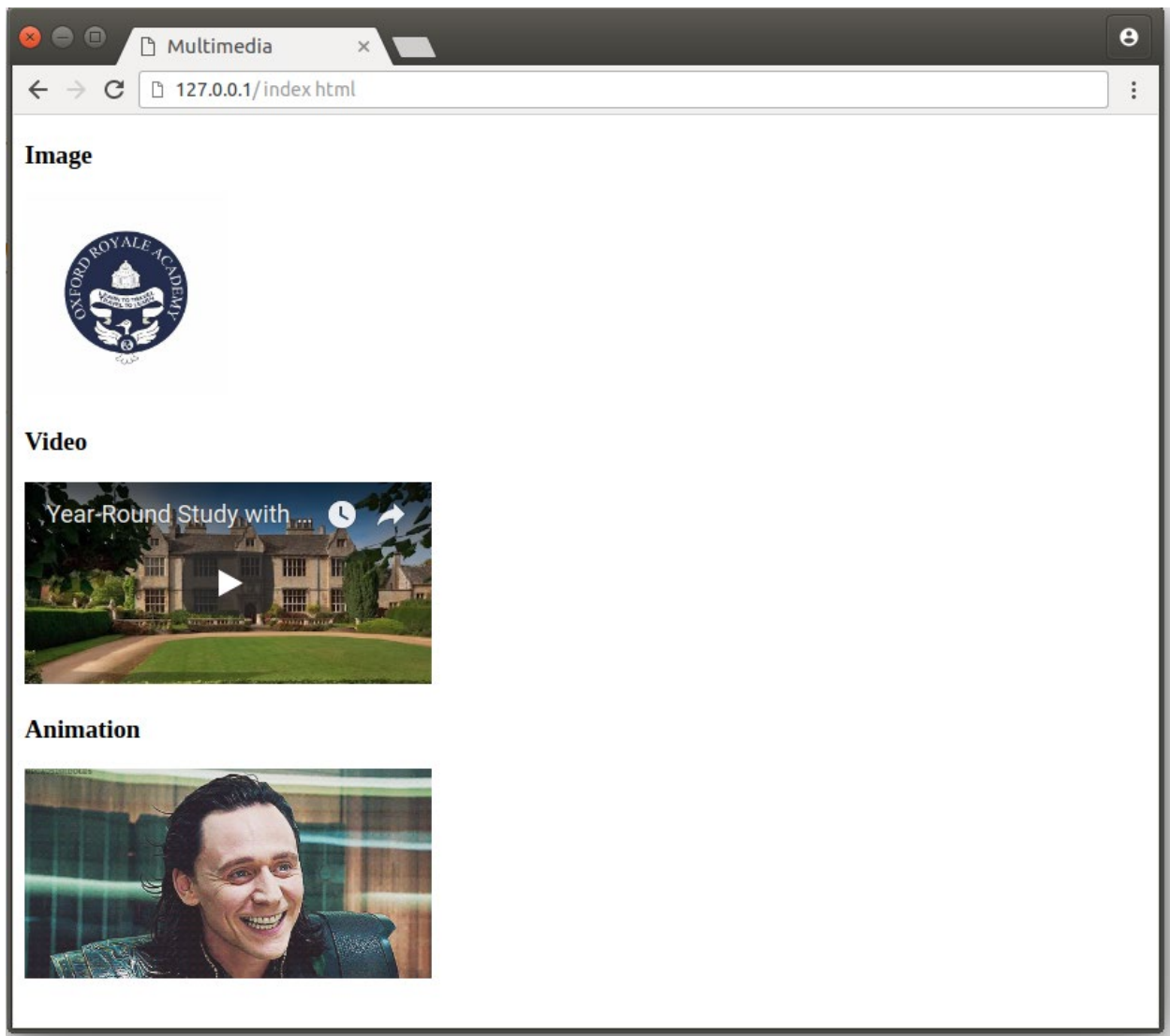
    <h3>Video</h3>
    <iframe width="300px" src="https://www.youtube.com/embed/XTtyhqSbnbw"
      frameborder="0" allow="autoplay; encrypted-media" allowfullscreen>
    </iframe>
    |
    <h3>Animation</h3>
    

  </body>
</html>
```

When you run your website, it will look as shown below.

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Lesson 5: CSS – web styling

Learning objectives:

- To learn how to define style attributes for items in the website
- To practise the different methods of adding CSS to a website

Lesson outline:

Styling items directly – inline CSS

Your teacher will:

- Demonstrate how to add style to an item using the style attribute
- Explain inline styling and why it holds the highest precedence compared to other styling methods

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Exercise 9

You have 10 minutes to:

- Add some inline styling to some tags in your website
- Change the text colour of a paragraph on your webpage to red or blue
- Change the font style to *italic*

Extra

- Add an inline style to the table cells created in Exercise 5
- Change the text size of a paragraph to 50px
- Change the background colour of your page

Adding internal CSS

Your teacher will:

- Explain CSS selectors for ids, classes, tags and so on
- Demonstrate how to add internal CSS in the head using the selectors explained

Exercise 10

You have 10 minutes to:

- Remove all inline CSS and put them in the head section of your page as internal CSS
- Set the background colour to “red” for elements with a “title” attribute and “blue” for elements with an “id” value

Adding external CSS

Your teacher will:

- Create a CSS file – style.css and populate it with content
- Demonstrate how to add an external CSS

No inline or internal CSS will be allowed in your final project – strictly external CSS

Exercise 11

You have 20 minutes to:

- Create your own CSS file with CSS font size, background colour and margin
- Import your CSS into your website
- Make sure that each page includes a reference to your external style sheet file
- Remove all other types of styling

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CSS – Web styling

Inline CSS example

```
<!DOCTYPE html>
<html>
  <head>
    <title> Inline CSS</title>
  </head>
  <body style="background-color: lightgray">

    <h1 style="color:red;">This is a heading</h1>
    <p style="color:darkgreen;">This is a paragraph.</p>

  </body>
</html>
```

Internal CSS example

```
<!DOCTYPE html>
<html>
  <head>
    <title> Internal CSS</title>
    <style>
      body {background-color: lightgray;}
      h1   {color: red;}
      p    {color: darkgreen;}
    </style>
  </head>
  <body>

    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>

  </body>
</html>
```

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External CSS example:

Create a CSS file named `styles.css` and save it in the same directory where your `html` file is saved. Below is the content you put in your `style.css`. You can add more styling later.

```
body {  
    background-color: lightgray;  
}  
h1 {  
    color: red;  
}  
p {  
    color: darkgreen;  
}
```

Attach the CSS file in your `HTML` file as shown.

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title> Inline CSS</title>  
    <link rel="stylesheet" href="styles.css">  
  </head>  
  <body>  
  
    <h1>This is a heading</h1>  
    <p>This is a paragraph.</p>  
  
  </body>  
</html>
```

When you run all three examples, your website will look as shown below.



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Lesson 6: HTML forms – User input

Learning objectives:

- To learn how to define forms in the website
- To practise the different input methods allowed in the HTML form tag

Lesson outline:

Introduction to forms

Your teacher will:

- Explain HTML forms and their importance for responsive websites
- Demonstrate creating an empty form

Adding form input elements

Your teacher will:

- Explain the different HTML5 input types – colour, date, datetime-local, email, month, number, range, search, tel time, URL week
- Demonstrate adding different form input elements including input-text, password, submit, reset, date, radio, checkbox, button

Exercise 12

You have 20 minutes to:

- Make a simple order form for customers who would like to purchase weekly, monthly, quarterly or annually subscriptions on your website. Add 4 input elements to ask for the user's first name, last name, address, city and email
- Consider the types of input element you might use for your project

Adding other HTML form elements

Your teacher will demonstrate adding other elements – select, textarea, label, button and datalist

Exercise 13

You have 10 minutes to:

- Add 2 datalists to your HTML form, one for "Country" (a list of 5-10 countries) and the other for the different types of subscription in Exercise 12. Add textarea and a "Submit Order" button at the end of your form. Think about how you can implement HTML forms in your project

HTML5 form attributes

Your teacher will explain the different HTML form attributes – autocomplete, autofocus, formnovalidate, min and max, multiple, list, height and width, placeholder, required, formaction, pattern.

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Exercise 14

You have 5 minutes to:

- Use a placeholder to add the text “First name” and “Last name” as a hint to your website user, so they know what can be entered in the input field
- Use the min and max attributes to set a restriction on users’ input in the text areas

Additional supporting materials

HTML forms – User input

Forms and inputs

The code below shows how to use forms and inputs.

```
<!DOCTYPE html>
<html>
  <head>
    <title> Forms and inputs</title>
  </head>
  <body>
    <form>
      <h3>Enter your details</h3>
      First name: <br>
      <input type="text" name="firstname"><br>
      Last name:<br>
      <input type="text" name="lastname"><br>
      Address:<br>
      <textarea name="message" rows="5" cols="30"></textarea><br>

      Gender:<br>
      <input type="radio" name="gender" value="male" checked> Male
      <input type="radio" name="gender" value="female"> Female
      <input type="radio" name="gender" value="other"> Other <br>

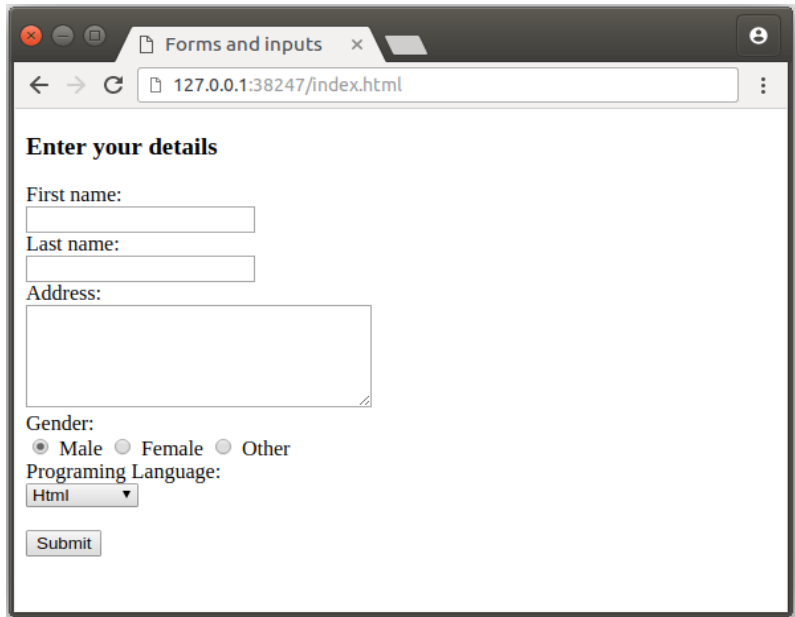
      Programing Language:<br>
      <select name="">
        <option value="html">Html</option>
        <option value="python">Python</option>
        <option value="java">Java</option>
        <option value="Javascript">JavaScript</option>
      </select>

      <br>
      <br>
      <button type="button">Submit</button>
    </form>
  </body>
</html>
```

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When you run the code, your website will look as shown below.



The screenshot shows a web browser window with the title 'Forms and inputs'. The address bar shows '127.0.0.1:38247/index.html'. The main content area displays a form titled 'Enter your details'. The form contains the following elements: 'First name:' with a text input field, 'Last name:' with a text input field, 'Address:' with a larger text input field, 'Gender:' with three radio buttons labeled 'Male', 'Female', and 'Other', 'Programming Language:' with a dropdown menu showing 'Html', and a 'Submit' button at the bottom.

Lesson 7: Web scripting – JavaScript

Learning objectives:

- To learn how to bring your website to life with JavaScript

Lesson outline:

Introduction to JavaScript

Your teacher will:

- Explain how JavaScript is used in modern websites
- Introduce the `<script>` tag and its placement position
- Explain the `<noscript>` tag and its importance

JavaScript to change HTML content

Your teacher will:

- Create an empty website with a paragraph
- Demonstrate using JavaScript how to change HTML content in that paragraph

Exercise 15

You have 10 minutes to:

- Create a blank website with the header “Adding JavaScript to HTML”. Then create a new paragraph with the text “JavaScript is a scripting language used to make your webpages more interactive” using JavaScript

Extra:

- What is the difference between a scripting language- and object-orientated programming language?

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- List 3 other popular examples of scripting language

JavaScript to change element style and attributes:

Your teacher will demonstrate using JavaScript to change the style and attributes of an element.

Exercise 16

You have 10 minutes to:

- Use JavaScript to change the style and attribute of the content in the paragraph you created in Exercise 15 (you can change the text, entirely or the text size to 40 pixels or add a blue background colour)

Extra:

- Find three websites and view the page source codes of the first page of each website to identify which of them used a script language such as JavaScript

JavaScript for output

Your teacher will:

- Show you how to change content using formulae and the innerHTML property
- Demonstrate using document.write() to write output to the document
- Demonstrate using document.getElementById() to write into an HTML element
- Demonstrate using window.alert() to give a pop-up message

Exercise 17

You have 10 minutes to:

- Use the document.write() to write a text or expression to a document
- Using document.getElementById() to write into an HTML element on your webpage
- Use window.alert() method in your JavaScript to create a pop-up alert message for your website

Extra

- Using JavaScript, create a webpage that accepts two numbers as user input and calculates addition and multiplication on numbers entered

Additional supporting materials

Web scripting – JavaScript

Change item style using JavaScript

Below is some sample JavaScript code that will change the style of the content in your paragraph with id “ora1” from plain to styled. There is also a button at the bottom of the page that, when clicked, activates/runs the JavaScript method changer to change the styling in your paragraph.

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```

<!DOCTYPE html>
<html>
  <body>

    <h1>My First JavaScript</h1>
    <p id="oral">Sample text to change style</p>
    <script>
      function changer() {
        document.getElementById("oral").style.fontSize = "25px";
        document.getElementById("oral").style.color = "red";
        document.getElementById("oral").style.backgroundColor = "yellow"; }
    </script>
    <button type="button" onclick="changer()">Click to change style</button>
  </body>
</html>

```

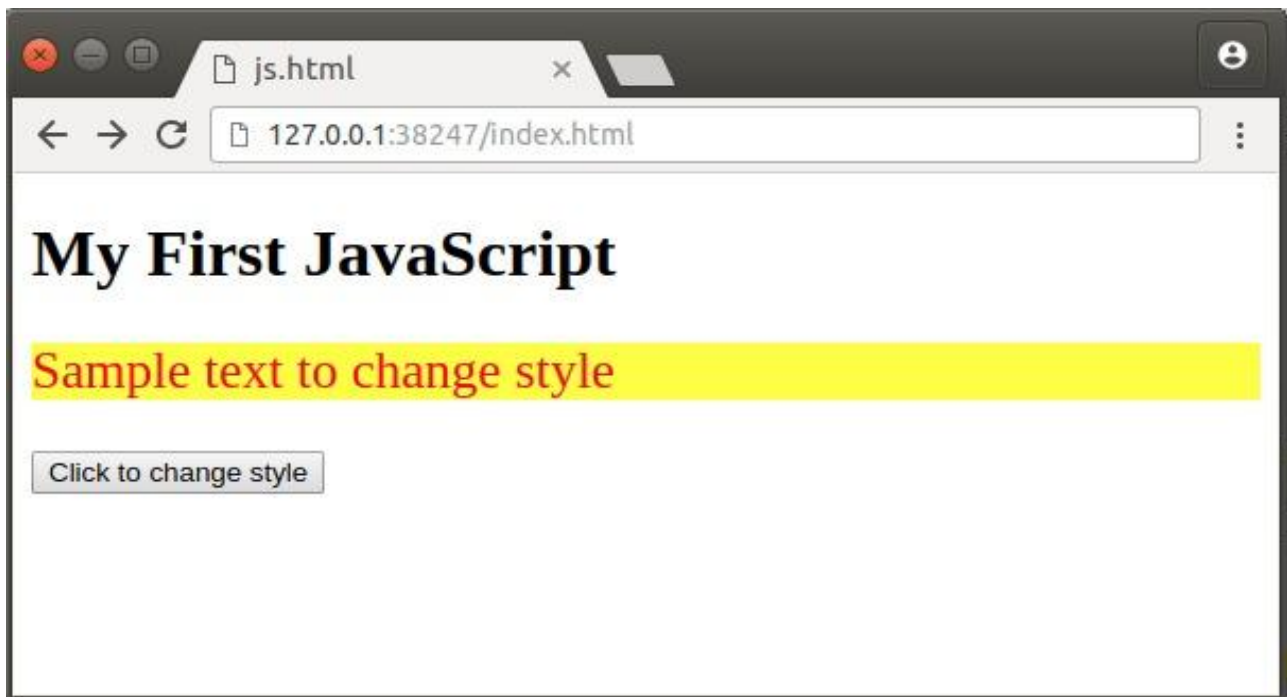
This is what your page would look like before clicking the button.



This is what it looks like after clicking the button.

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Exercise 17 extra task example:

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Sample form :

1st Number : 12

2nd Number: 10

Multiply

Divide

The Result Is :

120

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Lesson 8: Lab 2 project

Learning objectives:

- To work on your web development project

Website development project outline

You are required to create a website selling a futuristic product which does not yet exist but which would be a great seller. The website should describe this product, how it works and why it would be such a bestseller. You should also include images of the said product, either hand-drawn or created with the help of CAD. There are two stages to this section of the project: formatting and markup. It is possible to do much of the work before finishing the lessons on web programming. For example, it is possible to decide on a page layout, font and colour scheme on paper independent of the methods to enforce them.

The webpage should incorporate the following features:

- At least one figure
- Formatted text
- At least one table or nested list
- Captions for all tables and figures

Your style sheet should obey basic principles of design:

- High contrast between the text and the background
- Some use of colour to add interest, *e.g.* on section headings
- Styled `<em>` and `<strong>` tags, not `<i>` and `<b>` tags

There are a number of bonus marks available for more challenging formatting, *e.g.*:

- A functional menu that is well styled
- Hyperlinks between footnotes and the citations in the text
- Hyperlinked cross-references to figures, tables, etc. These do not need to be back-referenced

Make some notes about your code if you need to and feel free to talk to your teacher if you have any questions.

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Lesson 9: Welcome to Python

Learning objectives:

- To understand the background of programming languages
- To get all the necessary installations done on the computers in preparation for the rest of the course
- To understand the Python programming assignment and presentation due in Wednesday of Week 2

Lesson outline:

Welcome to Python

Your teacher will briefly explain the history of programming languages and the events that eventually led to languages like Python being developed.

Python programming assignment

Your teacher will:

- Explain what you have to do for the Python programming assignment due in by Wednesday of Week 2

If you are unclear about the assignment requirements, make sure to ask any questions you have now.

Get the tools:

- Download and install Python 3 from <https://www.python.org/downloads/>
- Download and install the Community edition of Pycharm or any other IDE of your choice that you will use to write Python code from <https://www.jetbrains.com/pycharm/download/>

First Python program demonstration

Your teacher will:

- Demonstrate how to create a new Python file
- Create a program that takes a name from the user and then greets the user

Exercise 18

You have 10 minutes to:

- Create your first Python program. Write a program that accepts a name as user input and prints a greeting addressed to the user with their name. Run your program to see the program output

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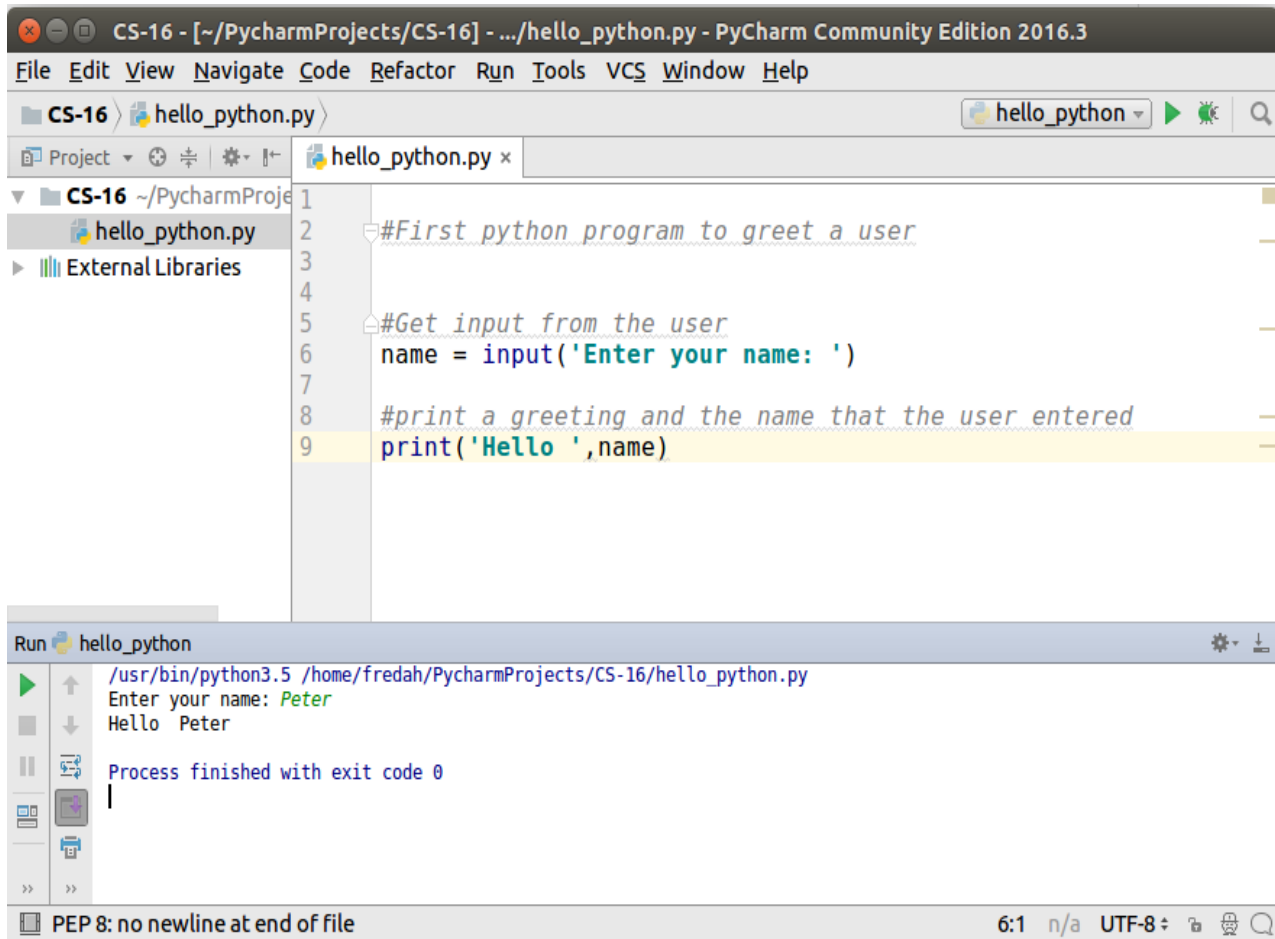
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Additional supporting materials

Welcome to Python

First Python program

Hello Python



The screenshot shows the PyCharm Community Edition 2016.3 interface. The main editor window displays a Python file named `hello_python.py` with the following code:

```
1 #First python program to greet a user
2
3
4
5 #Get input from the user
6 name = input('Enter your name: ')
7
8 #print a greeting and the name that the user entered
9 print('Hello ',name)
```

The code is executed, and the Run window shows the output:

```
Run hello_python
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/hello_python.py
Enter your name: Peter
Hello Peter
Process finished with exit code 0
```

The status bar at the bottom indicates a PEP 8 error: "PEP 8: no newline at end of file".

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Lesson 10: Variables and operators

Learning objectives:

- To learn the rules to be followed when creating and using variables
- To understand the different operators and their usage

Lesson outline:

Variables and data types

Your teacher will:

- Introduce variables with the naming conventions for variables and reserved words that are unavailable for variable naming
- Elaborate on data types – integers (int), strings (str), float – and show you how to convert from one datatype to another

Exercise 19

You have 10 minutes to:

- Create 5 different variables of different data types
- Use type casting to change between the data types of the variables you created
- Use the `type()` method to confirm the type change

Extra

- Use the `int()` built-in function to convert "a" into an integer type
- Print the data type of "a" to check it worked
- Now try running the following line of code: `print a + " squared equals " + b + " squared " + c + " squared"`
- You should get an error – do you know why?
- Try using the built-in `str()` (convert to string) function to fix the error

Arithmetic operators

Your teacher will:

- Discuss the arithmetic operators in your guides
- Discuss operator precedence for arithmetic operations in an expression
- Discuss integer division in Python

Exercise 20

You have 5 minutes to try out using different operators in expressions:

- Write a Python program that accepts two integers and performs arithmetic operations such as addition, subtraction, multiplication and division on the integers
- Try using the arithmetic operators on floating-point numbers

Extra

- Create a program that asks a user to enter their name and age. Your program should

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calculate the year when they will turn 100 years old and display the result as a message to the user

Comparison operators

Your teacher will:

- Discuss the different comparison operators
- Demonstrate their use

Exercise 21

You have 10 minutes to work with at least 3 different comparison operators:

- Write a Python program that accepts two integers as its input and compares the integers using the comparison operators (`>`, `<`, `==`, `!=`, `<=`, `>=`). Run your program to display your output

Logical operators

Your teacher will discuss the logical operators AND, OR, NOT.

Assignment operators

Your teacher will:

- Discuss the different assignment operators
- Demonstrate their use

Exercise 22

You have 5 minutes to experiment with at least 3 different assignment and logical operators:

- Apply logical operators to your comparisons in Exercise 21
- Write another Python program that accepts two integers as its input. Apply these three assignment operators (`+=`, `-=`, `/=`) to your integers. Run your program to display your output

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Variables and operators

Python reserved words

Keywords in Python programming language cannot be used as variable names. Some are listed below.

False	class	finally	is	return
None	continue	for	lambda	try
True	def	from	nonlocal	while
and	del	global	not	with
as	elif	if	or	yield
assert	else	import	pass	
break	except	in	raise	

Arithmetic operators in Python

Operator	Meaning	Example
+	Add two operands or unary plus	x + y +2
-	Subtract right operand from the left or unary minus	x - y -2
*	Multiply two operands	x * y
/	Divide left operand by the right one (always results in float)	x / y
%	Modulus – remainder of the division of left operand by the right	x % y (remainder of x/y)
//	Floor division – division that results into whole number adjusted to the left in the number line	x // y

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**	Exponent – left operand raised to the power of right	x**y (x to the power y)
----	--	----------------------------

Arithmetic operators example

The screenshot shows the PyCharm IDE with a file named `operators.py` open. The code defines two variables, `x = 13` and `y = 5`, and then prints the results of various arithmetic operations. The output window at the bottom shows the execution results.

```

1  x = 13
2  y = 5
3
4  # Output: x + y = 18
5  print('x + y =', x + y)
6
7  # Output: x - y = 8
8  print('x - y =', x - y)
9
10 # Output: x * y = 65
11 print('x * y =', x * y)
12
13 # Output: x / y = 2.6
14 print('x / y =', x / y)
15
16 # Output: x // y = 2
17 print('x // y =', x // y)
18
19 # Output: x ** y = 371293
20 print('x ** y =', x ** y)
21

```

Run operators
 /usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/operators.py
 x + y = 18
 x - y = 8
 x * y = 65
 x / y = 2.6
 x // y = 2
 x ** y = 371293
 Process finished with exit code 0

Comparison operators in Python

Operator	Meaning	Example
>	Greater than – True if left operand is greater than the right	x > y
<	Less than – True if left operand is less than the right	x < y
==	Equal to – True if both operands are equal	x == y
!=	Not equal to – True if operands are not equal	x != y
>=	Greater than or equal to – True if left operand is greater than or equal to the right	x >= y

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Comparison operators example

The screenshot shows the PyCharm Community Edition 2016.3 interface. The main editor displays a file named `comparisons.py` with the following code:

```
1 x = 20
2 y = 25
3
4 # Output: x > y is False
5 print('x > y is',x>y)
6
7 # Output: x < y is True
8 print('x < y is',x<y)
9
10 # Output: x == y is False
11 print('x == y is',x==y)
12
13 # Output: x != y is True
14 print('x != y is',x!=y)
15
16 # Output: x >= y is False
17 print('x >= y is',x>=y)
18
19 # Output: x <= y is True
20 print('x <= y is',x<=y)
```

The Run tool window at the bottom shows the execution output:

```
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/comparisons.py
x > y is False
x < y is True
x == y is False
x != y is True
x >= y is False
x <= y is True
Process finished with exit code 0
```

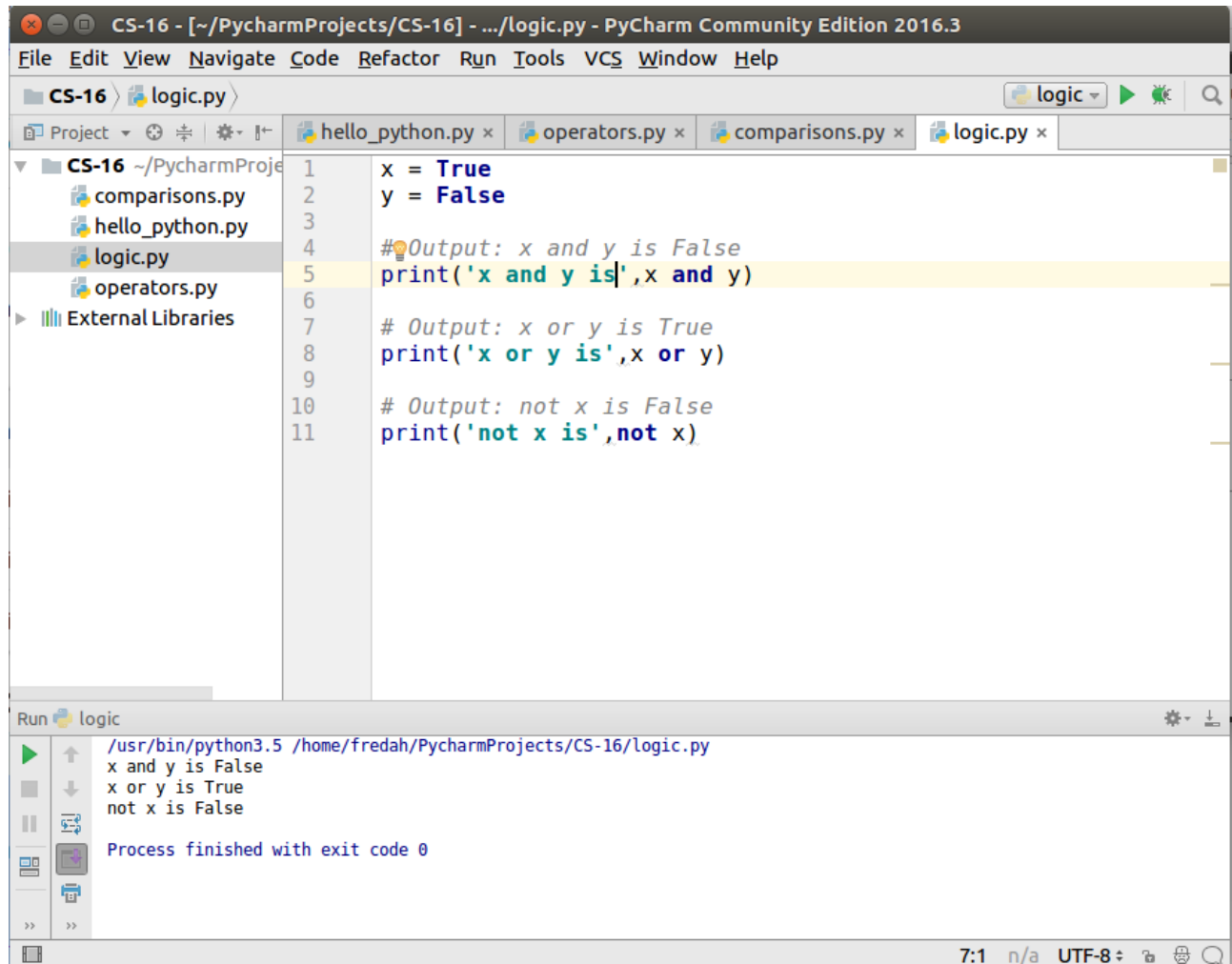
Logical operators in Python

Operator	Meaning	Example
and	True if both the operands are true	x and y
or	True if either of the operands is true	x or y
not	True if the operand is false (complements the operand)	not x

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Logical operators example



The screenshot shows the PyCharm IDE interface. The top toolbar includes menus for File, Edit, View, Navigate, Code, Refactor, Run, Tools, VCS, Window, and Help. The project browser on the left shows a project named 'CS-16' with files 'comparisons.py', 'hello_python.py', 'logic.py', and 'operators.py'. The main editor displays the content of 'logic.py' with the following code:

```
1 x = True
2 y = False
3
4 #Output: x and y is False
5 print('x and y is', x and y)
6
7 # Output: x or y is True
8 print('x or y is', x or y)
9
10 # Output: not x is False
11 print('not x is', not x)
```

Below the editor is the 'Run' window, which shows the command executed: `/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/logic.py`. The output of the program is displayed as:

```
x and y is False
x or y is True
not x is False
```

The status bar at the bottom indicates the current line is 7:1, the encoding is n/a, and the file is in UTF-8.

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Assignment operators in Python

Operator	Example	Equivalent to
=	x = 5	x = 5
+=	x += 5	x = x + 5
-=	x -= 5	x = x - 5
*=	x *= 5	x = x * 5
/=	x /= 5	x = x / 5
%=	x %= 5	x = x % 5
//=	x //= 5	x = x // 5
**=	x **= 5	x = x ** 5
&=	x &= 5	x = x & 5
=	x = 5	x = x 5
^=	x ^= 5	x = x ^ 5
>>=	x >>= 5	x = x >> 5
<<=	x <<= 5	x = x << 5

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Identity and membership operators

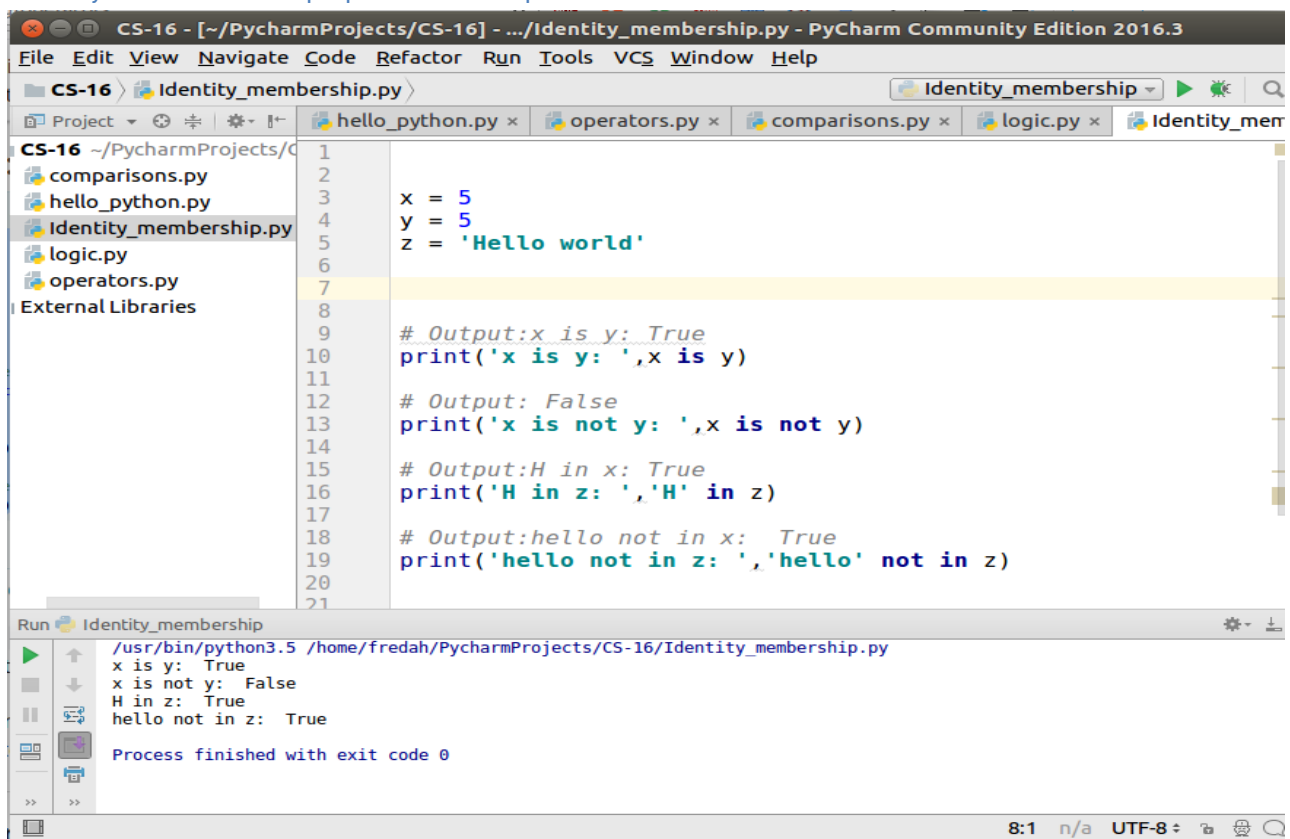
Identity operators

Operator	Meaning	Example
is	True if the operands are identical (refer to the same object)	x is True
is not	True if the operands are not identical (do not refer to the same object)	x is not True

Membership operators

Operator	Meaning	Example
in	True if value/variable is found in the sequence	H in Hello
not in	True if value/variable is not found in the sequence	p not in Hello

Identity and membership operators example



The screenshot shows the PyCharm IDE with a project named 'CS-16'. The file 'Identity_membership.py' is open and contains the following code:

```
1 x = 5
2 y = 5
3 z = 'Hello world'
4
5
6
7
8
9 # Output: x is y: True
10 print('x is y: ', x is y)
11
12 # Output: False
13 print('x is not y: ', x is not y)
14
15 # Output: H in x: True
16 print('H in z: ', 'H' in z)
17
18 # Output: hello not in x: True
19 print('hello not in z: ', 'hello' not in z)
20
21
```

The Run window at the bottom shows the output of the script:

```
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/Identity_membership.py
x is y: True
x is not y: False
H in z: True
hello not in z: True
Process finished with exit code 0
```

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Lesson 11: Control statements

Learning objectives:

- To learn how to use control statements to enforce control flow in your Python code
- To understand the difference between if, if-else and if-elif-else

Lesson outline:

If statement

Your teacher will:

- Introduce control flow using a flow chart to help illustrate the importance of if statements
- Explain the syntax of an if statement and demonstrate how to use it

Exercise 23

You have 10 minutes to:

- Write a Python program that accepts an integer as temperature and creates an if statement that reports if the temperature is hot

Extra

- Write a Python program that accepts the age of a student as input and displays the output “Your age is out of range for this class” if the student’s age is greater than or equal to 18

If-else statement

Your teacher will:

- Explain the need for an else statement in if statements
- Demonstrate how to create an if-else statement

Exercise 24

You have 5 minutes to:

- Modify your program in Exercise 23 to display an output if the temperature is either hot or cold

Extra

- Write a Python program that sets an age limit of at least 18 to users who would like to purchase a car. The user is only eligible to buy a car if they are 18 years old and above

If-elif-else statement

Your teacher will:

- Demonstrate an if-elif-else statement
- Explain why you are allowed to have as many elifs as possible and only one else

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Exercise 25

You have 10 minutes to:

- Modify the program in Exercise 24 to report if the temperature is hot, cold or warm

Extra

- Write a Python program that accepts temperature as an input and converts the temperature from Celsius to Fahrenheit and Fahrenheit to Celsius [Hint: Fahrenheit = ((Celsius temperature * 9)/5) + 32 AND Celsius = (Fahrenheit temperature - 32) * 5/9]
- Write a program that accepts a number as a user input and checks if the number is a prime, even or odd number. Display a message with the result as output

Nested if statements

Your teacher will use the BMI calculator example to show you how nested if statements could be used.

Exercise 26

You have 10 minutes to:

- Write a Python program that accepts integers from 1 to 7 and prints their corresponding day of the week. That is, when the user enters the number 1, your program should display “Sunday”; when the user enters the number 2, “Monday” should be displayed, and so on. Your program should also give an “Invalid number” response if the user enters any other integer. This task will require nested if statements and possibly a combination of if, elif and else as well.

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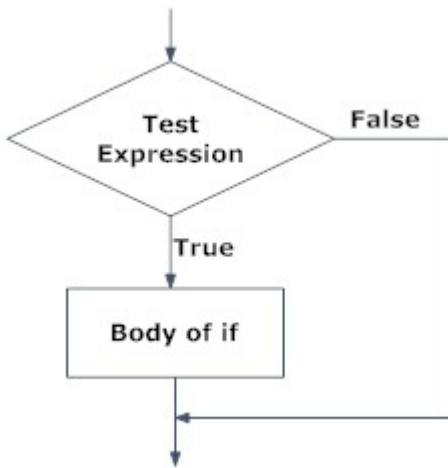
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Additional supporting materials

Control statements

If statement

If statement flowchart



If statement example

The screenshot shows the PyCharm IDE with a project named "CS-16". The file explorer on the left lists several Python files, including "if.py". The main editor window displays the code in "if.py":

```
1 # If the xr is positive, we print an appropriate message
2
3
4 x = 7
5 if x > 0:
6     print(x, "is a positive number.")
7     print("This is always printed.")
8
9 y = -19
10 if y > 0:
11     print(y, "is a positive number.")
12
13 print("This is also always printed.")
```

Below the editor, the "Run" window shows the output of the program:

```
Run if
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/if.py
7 is a positive number.
This is always printed.
This is also always printed.
Process finished with exit code 0
```

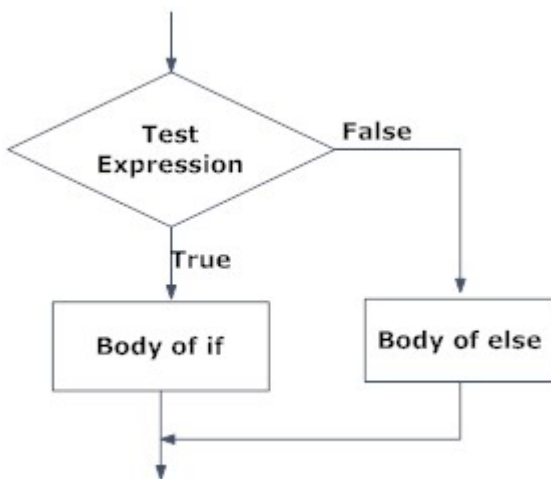
The status bar at the bottom indicates the time is 12:5, the encoding is n/a, and the character set is UTF-8.

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If-else statements

If-else statement flowchart



If-else statement example

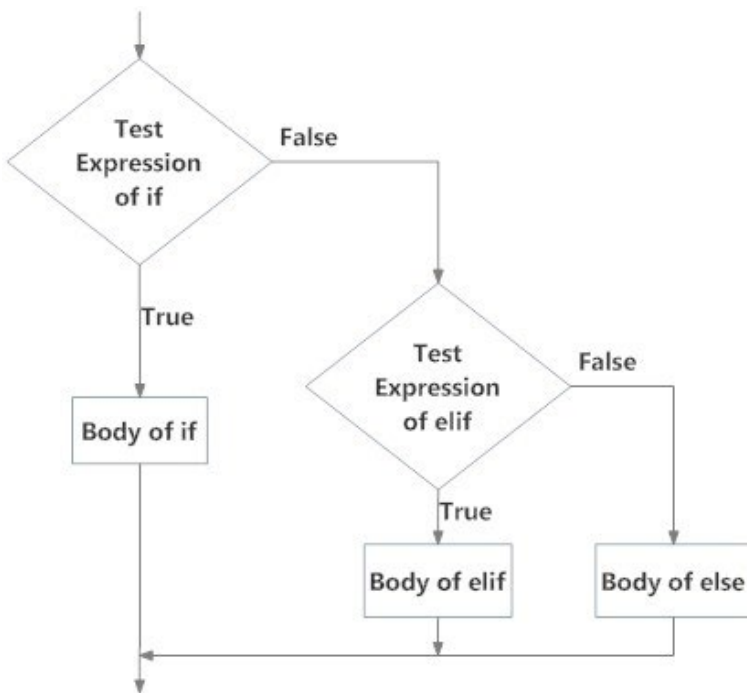
```
CS-16 - [~/PycharmProjects/CS-16] - .../if_else.py - PyCharm Community Edition 2016.3
File Edit View Navigate Code Refactor Run Tools VCS Window Help
CS-16 > if_else.py
Project | erators.py x | comparisons.py x | logic.py x | Identity_membership.py x | if.py x | if_else.py
CS-16 ~/PycharmProjects/CS-16
1 # Program checks if the number is positive or negative
2 # And displays an appropriate message
3
4
5 x = 50
6
7 # Try these two variations as well.
8 # x = -5
9 # x = 0
10
11 if x >= 0:
12     print("Positive or Zero")
13 else:
14     print("Negative number")
Run if_else
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/if_else.py
Positive or Zero
Process finished with exit code 0
5:1 n/a UTF-8
```

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If-elif-else statements

If-elif-else statement flowchart



If-elif-else example

```
1  # In this program we get PH input from the user,
2  # check if the ph is acidic, basic, neutral or out of range
3  # display an appropriate message
4
5  ph = float(input('Enter PH: '))
6
7  # Try these other variations as well:
8  # ph = 7.6
9  # ph = -39.6
10 # ph = 9
11
12 if ph >= 0 and ph <= 6:
13     print("Acidic")
14 elif ph > 6 and ph < 8:
15     print("Neutral")
16 elif ph >= 8 and ph <= 14:
17     print("Basic")
18 else:
19     print("Out of range")
20
21
```

Run elif

```
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/elif.py
Enter PH: 5.3
Acidic
Process finished with exit code 0
```

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Nested if statements

Example – Health risk

The table below shows the risk level in relation to the age and BMI of an individual. Here we see how nested if statements can determine, given the age, weight and height, the risk level of the individual.

		Age	
		< 45	>=45
BMI	< 22	Low	Medium
	>=22	Medium	High

```
CS-16 - [~/PycharmProjects/CS-16] - .../nested_if.py - PyCharm Community Edition 2016.3
File Edit View Navigate Code Refactor Run Tools VCS Window Help
CS-16 > nested_if.py >
Project > Identity_membership.py x if.py x if_else.py x elif.py x nested_if.py x
CS-16 ~/PycharmProjects/CS-16
1 # Example if nested if statement using the bmi calculator
2
3 age = float(input('Enter your age: '))
4 weight = float(input('Enter your weight(kg): '))
5 height = float(input('Enter your height(m): '))
6 bmi = weight/height**2
7 print('BMI: ',bmi)
8
9 if age < 45:
10     if bmi < 22.0:
11         risk = 'low'
12     else:
13         risk = 'medium'
14 else:
15     if bmi < 22.0:
16         risk = 'medium'
17     else:
18         risk = 'high'
19 print('The risk is: ',risk)
20
21

Run: nested_if nested_if
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/nested_if.py
Enter your age: 17
Enter your weight(kg): 68
Enter your height(m): 1.4
BMI: 34.693877551020414
The risk is: medium
Process finished with exit code 0
9:1 n/a UTF-8
```

Extra:

How can you improve the given solution for the BMI calculator? There are several ways this can be done, so give it a try.

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Lesson 12: Iterations and conditions

Learning objectives:

- To learn how to use conditional statements to enforce control flow in your Python code
- To understand the difference between for and while loops

Lesson outline:

For loop

Your teacher will:

- Introduce iteration using a flow chart to help illustrate the flow in a for loop
- Explain the syntax of a for loop and demonstrate how to use it

Exercise 27

You have 20 minutes to:

- Given a list temp = [12, -27, 23.5, 46.8, 26.6, 17, 89, -32.6, 25], write a Python program using For loop that displays the maximum and minimum temperatures from the list

Your output should be:

max = 89; min = -32.6

Extra:

- Using For loop, write a Python program that displays the following pattern:

```
*
* *
* * *
* * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

While loop

Your teacher will:

- Introduce the while loop using a flow chart to help illustrate the flow in a while loop
- Explain the syntax of a while loop and demonstrate how to use it

Exercise 28

You have 20 minutes to:

- Write a Python program to find prime numbers between 100 and 150

Your program should output: 101, 103, 107, 109, 113, 127, 131, 137, 139, 149

Extra

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- Write a Python program that prints “I enjoy programming in Python” 10 times

Break and continue statements

Your teacher will:

- Introduce the break and the continue statements
- Explain their importance
- Demonstrate how either can be used within a loop

Extra

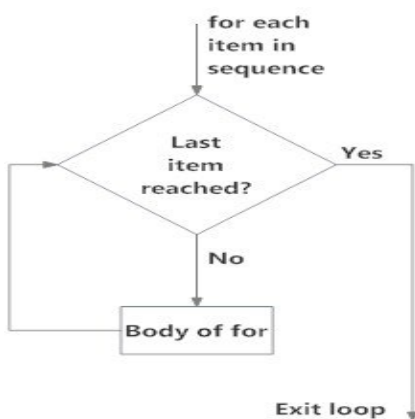
- Using the continue statements, write a Python program that displays numbers from 0 to 10 except the numbers 5 and 9
- Write a Python program that accepts 10 integers as a tuple and prints the sum of the first 5 numbers in your tuple using the break statement

Additional supporting materials

Iterations and conditions

For loop

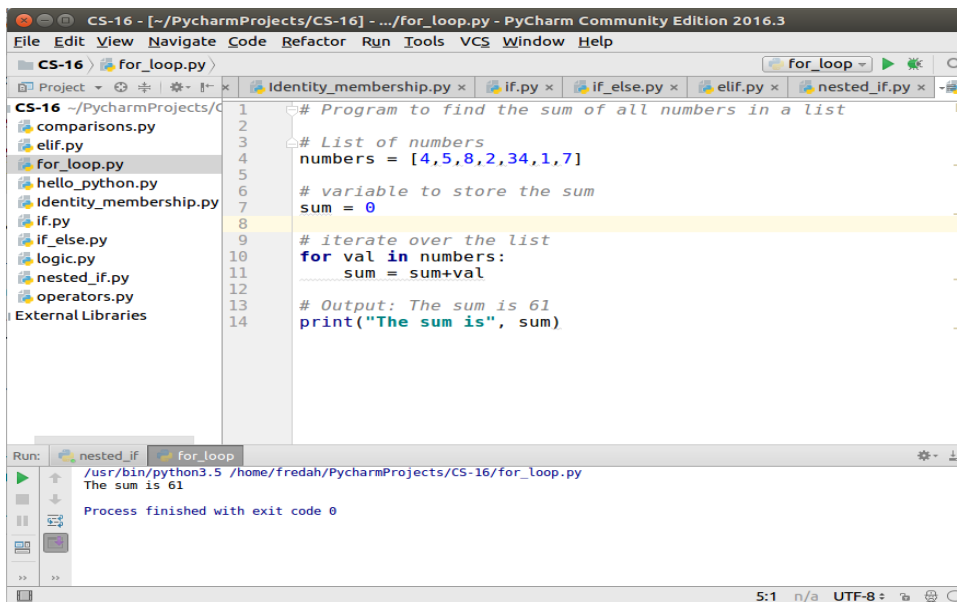
For loop flowchart



For loop example

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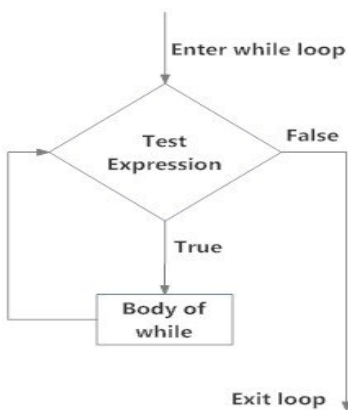


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While loop

While loop flowchart



While loop example

The screenshot shows the PyCharm IDE with a project named 'CS-16'. The file explorer on the left lists several Python files, including 'while.py'. The main editor window displays the code in 'while.py':

```
1  #This program print old numbers below 10
2
3  i = 0
4
5  while i <= 10:
6
7      #Check if a i is divisible by 2
8      if i%2 != 0:
9          print(i)
10     i += 1
```

The 'while' statement on line 5 is highlighted. Below the editor, the 'Run' console shows the output of the program:

```
1
3
5
7
9
```

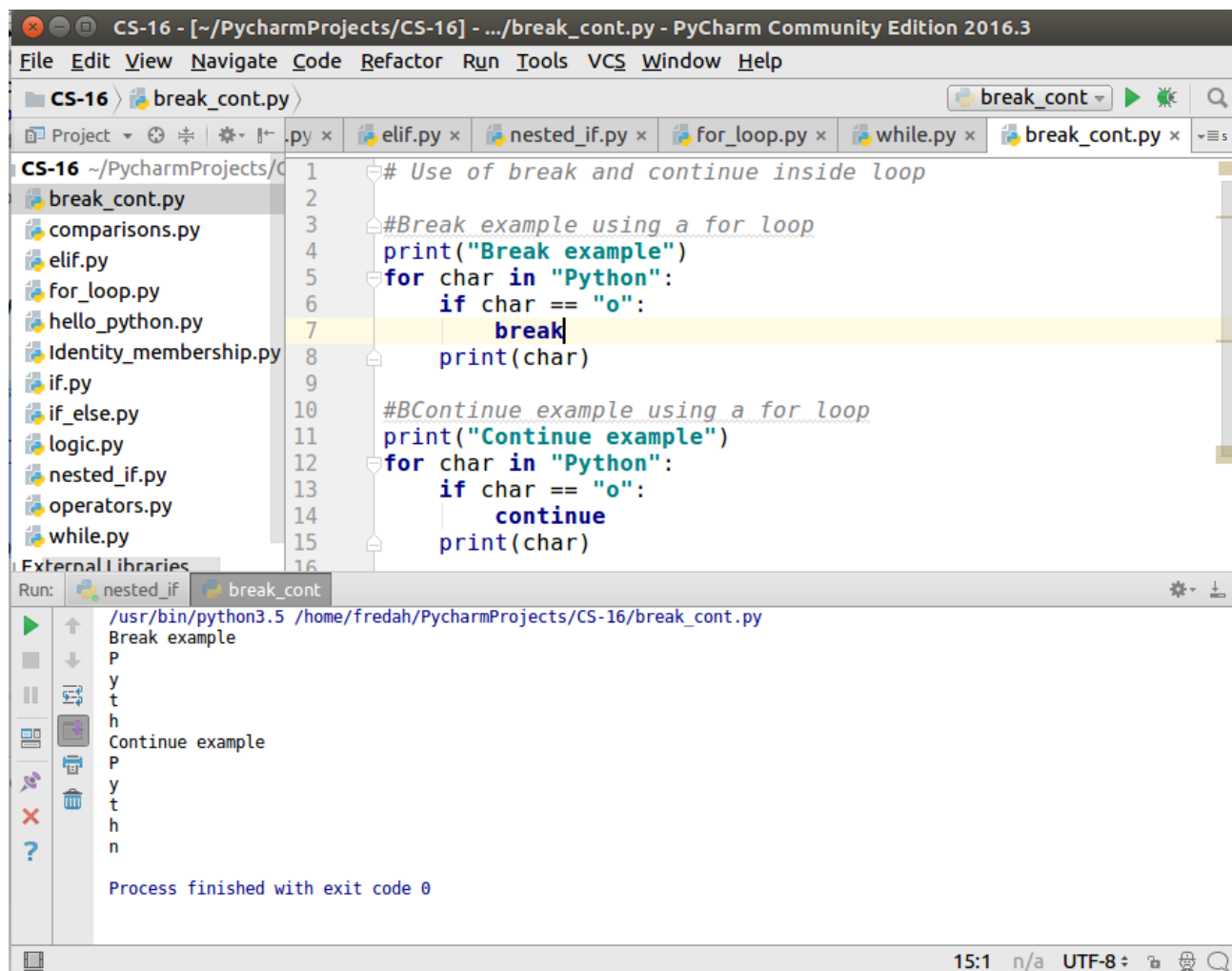
Below the output, it states 'Process finished with exit code 0'. The status bar at the bottom indicates the file is '9:1 n/a UTF-8'.

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Break and continue

Below is an example of how to use break and continue.



The screenshot shows the PyCharm IDE interface. The main editor window displays a Python file named `break_cont.py` with the following code:

```
1 # Use of break and continue inside loop
2
3 #Break example using a for loop
4 print("Break example")
5 for char in "Python":
6     if char == "o":
7         break
8     print(char)
9
10 #Continue example using a for loop
11 print("Continue example")
12 for char in "Python":
13     if char == "o":
14         continue
15     print(char)
16
```

The `break` statement is highlighted on line 7. The `continue` statement is highlighted on line 14. The `Run` button is visible in the top right corner of the editor window.

The `Run` console at the bottom shows the output of the script:

```
Run: /usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/break_cont.py
Break example
P
y
t
h
Continue example
P
y
t
h
n
Process finished with exit code 0
```

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Lesson 13: Lab 3 project

Learning objectives:

- To work on your projects

Web development project is due today!

Major project using Python

You are required to propose your own project; however, it has to be approved by the teacher.

Examples of projects you can work on are:

- Currency Converter – a program that accepts an amount in one currency from the user and converts it into any other currency on your list based on the user's choice. You will need a minimum of 10 currencies.
- Another example is to program a smart fridge. The smart fridge should know how old the eggs and bacon in it are. If you know that eggs spoil after 3 weeks (21 days) and bacon after one week (7 days), given the ageOfBacon and ageOfEggs in days, determine if you can cook bacon and eggs or what ingredient you need to throw out (this means, your smart fridge must keep track of when the bacon and eggs were placed in the fridge). If the bacon and eggs are still below their ages, display "You can cook the bacon and eggs". If you need to throw out any of the ingredients, for each one display a line that indicates which one should be thrown out – "Throw out <ingredient> – (where <ingredient> is bacon or eggs) in any order.

Please feel free to create your own project. The examples above are meant to guide your ideas.

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Lesson 14: Methods and functions

Learning objectives:

- To learn how to use and define your own methods and functions
- To learn how to call methods to run, pass arguments and return results

Lesson outline:

Functions

Your teacher will:

- Introduce functions
- Explain built-in functions and their uses
- Demonstrate how to use a built-in function
- Explain how to define your own function
- Demonstrate using their own function

Exercise 29

You have 20 minutes to:

- Write your own function that takes three integers as arguments, compares them and displays the largest integer

Extra

- Write a function that accepts two integers as arguments and displays the average of the two integers

Return statement

Your teacher will:

- Explain how the return statement works
- Demonstrate using a return statement in one of their methods
- Show how you process the result is given back to you by the method

Exercise 30

You have 20 minutes to:

- Write a Python program that calls the function you created in Exercise 29 to display the largest of the following numbers:
-10, 20, 30
-55, 40, 22
- Show that the result returned by the function is used after the method call

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Scope and lifetime of variables

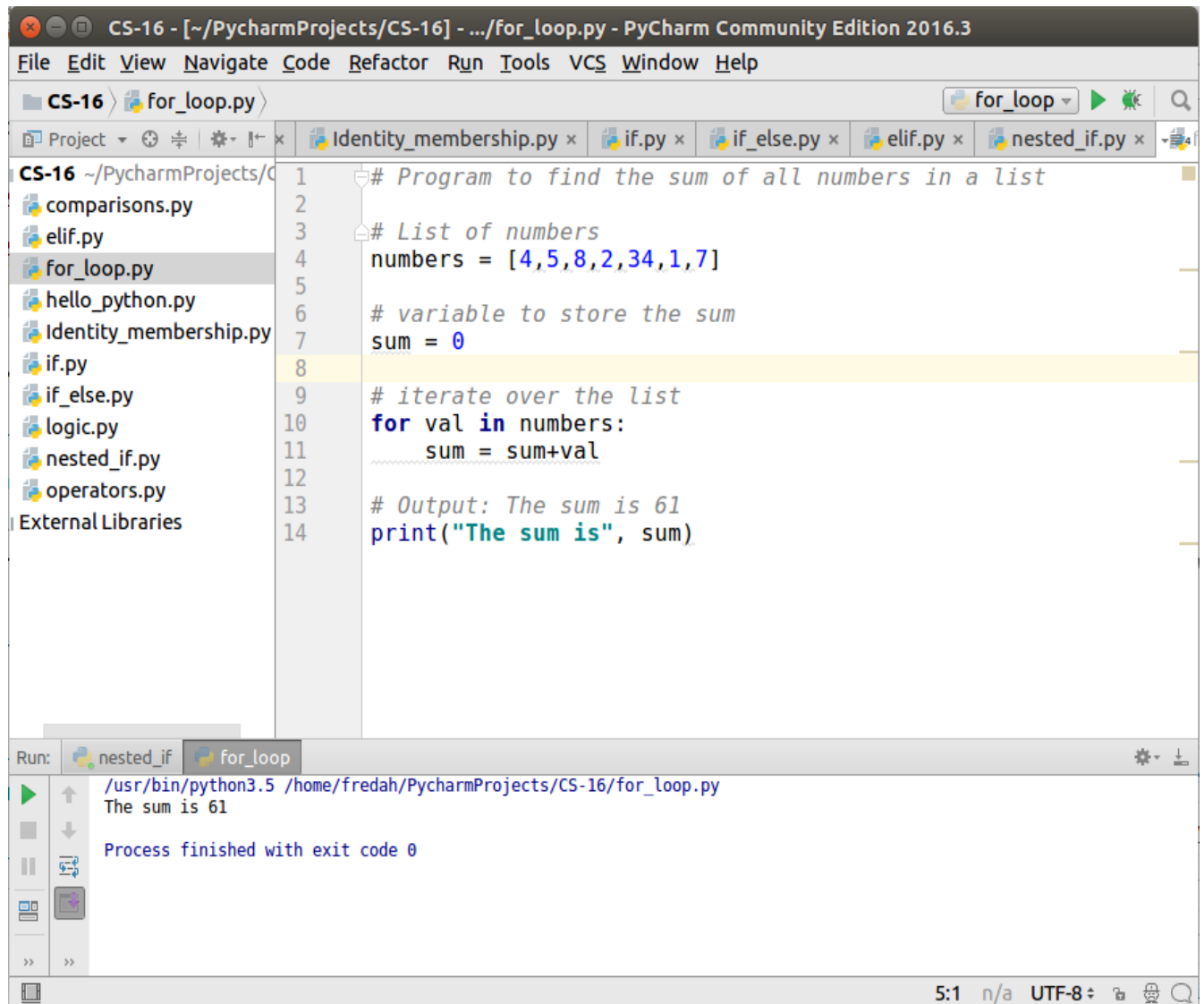
Your teacher will:

- Discuss scoping and how this affects variables defined within a certain scope like a method
- Explain the use of global in a method when you want to make changes to global variables

Additional supporting materials

Methods and functions

Functions example 1



The screenshot shows the PyCharm Community Edition 2016.3 interface. The main editor window displays a Python file named `for_loop.py` with the following code:

```
1  # Program to find the sum of all numbers in a list
2
3  # List of numbers
4  numbers = [4,5,8,2,34,1,7]
5
6  # variable to store the sum
7  sum = 0
8
9  # iterate over the list
10 for val in numbers:
11     sum = sum+val
12
13 # Output: The sum is 61
14 print("The sum is", sum)
```

The left sidebar shows the project structure for `CS-16`, with files including `comparisons.py`, `elif.py`, `for_loop.py` (selected), `hello_python.py`, `Identity_membership.py`, `if.py`, `if_else.py`, `logic.py`, `nested_if.py`, `operators.py`, and `External Libraries`.

The bottom panel shows the output of the program:

```
Run: /usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/for_loop.py
The sum is 61
Process finished with exit code 0
```

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Functions example 2

The screenshot shows the PyCharm IDE interface. The main editor window displays a Python file named `functions.py` with the following code:

```
1 #function examples
2 from math import sqrt
3
4 #Built in function
5 x = sqrt(100)
6 print('Square root is: ', x)
7
8 #user defined functions
9 def even(number):
10     """This function checks if a number is even"""
11     if number%2 == 0:
12         print(number, ' is even')
13     else:
14         print(number, ' is not even')
15
16 def calculate_bmi(age,height,weight):
17     """ This function takes in age,height,weight
18     and calculates the bmi and determine health risk"""
19     bmi = weight / height ** 2
20
21     if age < 45:
22         if bmi < 22.0:
23             risk = 'low'
24         else:
25             risk = 'medium'
26     else:
27         if bmi < 22.0:
28             risk = 'medium'
29         else:
30             risk = 'high'
31     return bmi,risk
32
33 #call methods
34 even(4)
35 print(calculate_bmi(17,1.6,60))
```

The left sidebar shows the project structure for `CS-16`, with `functions.py` selected. The bottom panel shows the output of the script:

```
Run: /usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/functions.py
Square root is: 10.0
4 is even
(23.437499999999996, 'medium')
```

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Lesson 15: Data structures

Learning objectives:

- To learn the different Python data structures and become comfortable with defining and modifying them
- To understand the difference between lists and dictionaries

Lesson outline:

Lists

Your teacher will:

- Define lists and show you how to create a list
- Demonstrate how to access elements in a list by means of list indexing and negative indexing
- Show you how to slice a list
- Show you how to add and remove elements in a list

Exercise 31

You have 20 minutes to:

- Write a Python program that, when given a list, should iterate over it with a For loop and count how many times each item appears in the list. That is, given the list below, your program should print how many times an element appears. For example:

```
list = [1,4,5,1,5,4,1,5,1,5,4,1]
```

Expected output:

1 appears 5 times

4 appears 3 times

5 appears 4 times

Dictionaries

Your teacher will:

- Define Python dictionaries and create one
- Demonstrate how to access elements in a dictionary
- Show how you can add and delete items from a dictionary
- Show how to iterate over a dictionary

Exercise 32

You have 20 minutes to:

- Modify your program in Exercise 31. This time you have to use a dictionary to keep track of how many times each item appears in the list. That is, given a list, your program prints how many times an element appears and uses a dictionary to keep track of your results for each

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element. For example:

```
list = [1,4,5,1,5,4,1,5,1,5,4,1]
```

Expected output:

```
{1:4, 4:3, 5:4}
```

Sets

Your teacher will:

- Introduce Python sets
- Demonstrate how to create, add or remove an element
- Demonstrate set operations – union, intersection, difference and symmetric difference

Extra

- Write a program that creates a set for the numbers [0,2,4,6,8,10]
- Use a For loop to iterate over the set and print each number in that set
- Add new numbers [12,14] to the set and print all the numbers in your set again to conform the new numbers have been added
- Remove the number [0] from your set
- Perform any of the set operations with another set of numbers [1,2,3,4,5,6,7]

Additional supporting materials

Data structures

Lists

Python list methods

append() Add an element to the end of the list

extend() Add all elements of a list to the another list

insert() Insert an item at the defined index

remove() Removes an item from the list

pop() Removes and returns an element at the given index

clear() Removes all items from the list

index() Returns the index of the first matched item

count() Returns the count of number of items passed as an argument

sort() Sort items in a list in ascending order

reverse() Reverse the order of items in the list

copy() Returns a shallow copy of the list

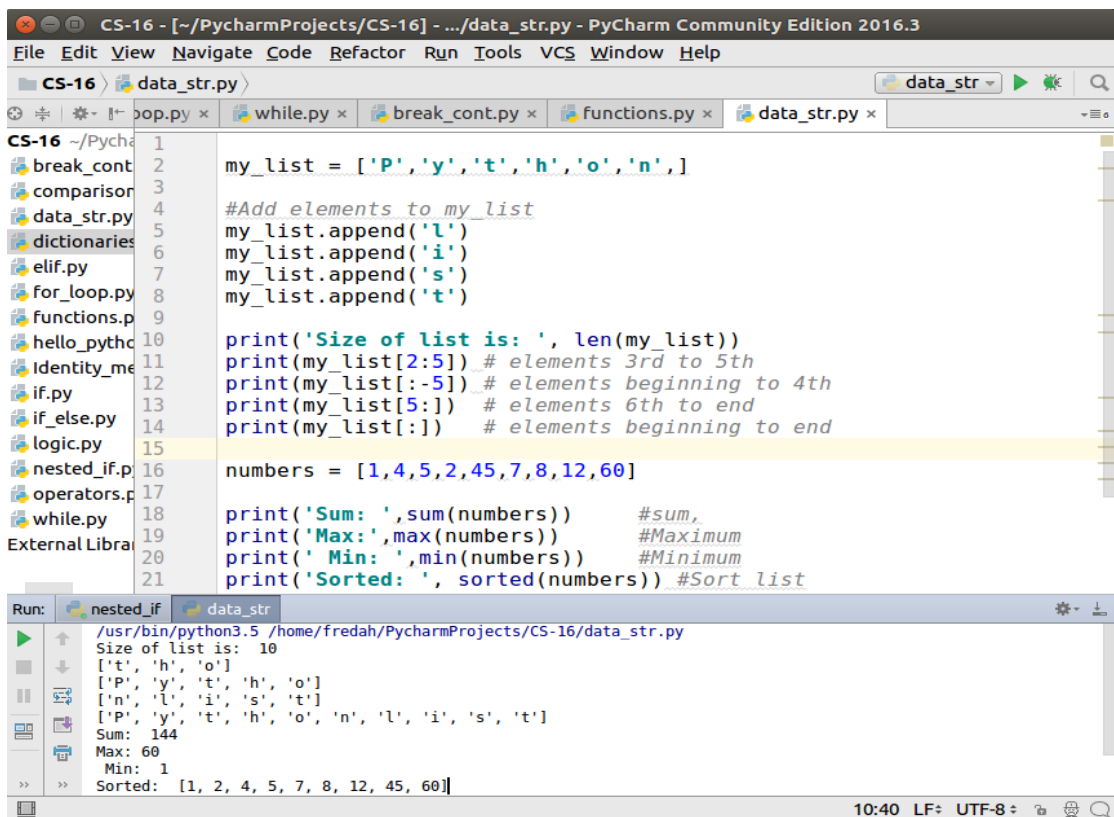
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Built-in functions with list

all()	Return True if all elements of the list are true (or if the list is empty)
any()	Return True if any element of the list is true. If the list is empty, return False
enumerate()	Return an enumerate object. It contains the index and value of all the items of list as a tuple
len()	Return the length (the number of items) in the list
list()	Convert an iterable (tuple, string, set, dictionary) to a list
max()	Return the largest item in the list
min()	Return the smallest item in the list
sorted()	Return a new sorted list (does not sort the list itself)
sum()	Return the sum of all elements in the list

Lists example



The screenshot shows the PyCharm IDE with a file named `data_str.py` open. The code defines a list `my_list` with elements `['P', 'y', 't', 'h', 'o', 'n']` and appends `'l', 'i', 's', 't'` to it. It then prints the size of the list (10) and slices of the list. Below this, a list of numbers `numbers = [1, 4, 5, 2, 45, 7, 8, 12, 60]` is defined, and the sum, maximum, minimum, and sorted version are printed. The Run window at the bottom shows the output of the script.

```
CS-16 - [~/PycharmProjects/CS-16] - .../data_str.py - PyCharm Community Edition 2016.3
File Edit View Navigate Code Refactor Run Tools VCS Window Help
CS-16 - data_str.py
data_str.py
1 my_list = ['P', 'y', 't', 'h', 'o', 'n', ]
2
3
4 #Add elements to my_list
5 my_list.append('l')
6 my_list.append('i')
7 my_list.append('s')
8 my_list.append('t')
9
10 print('Size of list is: ', len(my_list))
11 print(my_list[2:5]) # elements 3rd to 5th
12 print(my_list[:5]) # elements beginning to 4th
13 print(my_list[5:]) # elements 6th to end
14 print(my_list[:]) # elements beginning to end
15
16 numbers = [1, 4, 5, 2, 45, 7, 8, 12, 60]
17
18 print('Sum: ', sum(numbers)) #sum,
19 print('Max: ', max(numbers)) #Maximum
20 print('Min: ', min(numbers)) #Minimum
21 print('Sorted: ', sorted(numbers)) #Sort list

Run: nested_if data_str
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/data_str.py
Size of list is: 10
['t', 'h', 'o']
['P', 'y', 't', 'h', 'o']
['n', 'l', 'i', 's', 't']
['P', 'y', 't', 'h', 'o', 'n', 'l', 'i', 's', 't']
Sum: 144
Max: 60
Min: 1
Sorted: [1, 2, 4, 5, 7, 8, 12, 45, 60]
```

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Dictionaries

Python dictionary methods

clear()	Remove all items from the dictionary
copy()	Return a shallow copy of the dictionary
fromkeys(seq[, v])	Return a new dictionary with keys from seq and value equal to v (defaults to None)
get(key[,d])	Return the value of key. If key does not exist, return d (defaults to None)
items()	Return a new view of the dictionary's items (key, value)
keys()	Return a new view of the dictionary's keys
pop(key[,d])	Remove the item with key and return its value or d if key is not found. If d is not provided and key is not found, raises KeyError
popitem()	Remove and return an arbitrary item (key, value). Raises KeyError if the dictionary is empty
setdefault(key[,d])	If key is in the dictionary, return its value. If not, insert key with a value of d and return d (defaults to None)
update([other])	Update the dictionary with the key/value pairs from other, overwriting existing keys
values()	Return a new view of the dictionary's values

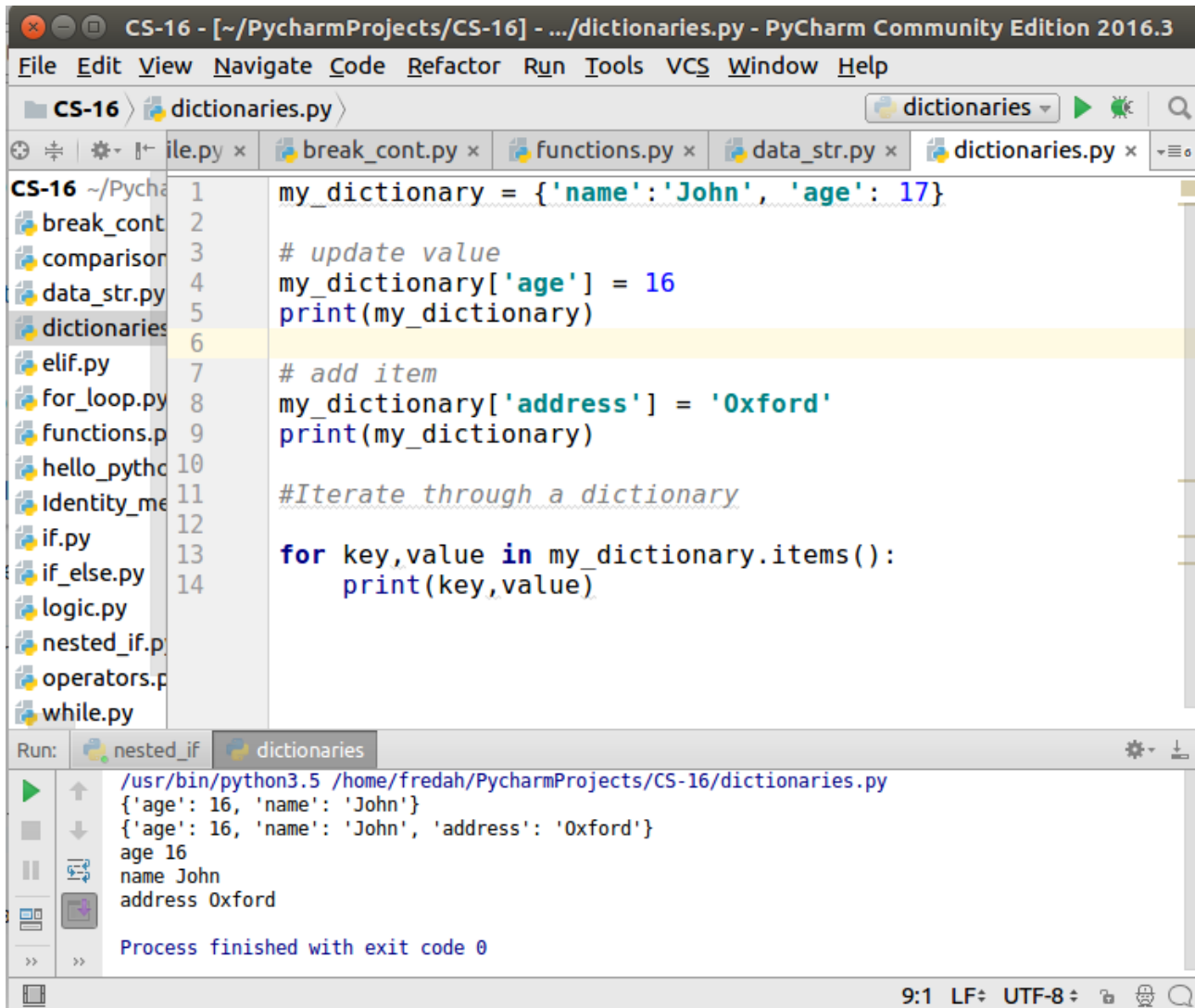
Built-in functions with dictionary

all()	Return True if all keys of the dictionary are true (or if the dictionary is empty)
any()	Return True if any key of the dictionary is true. If the dictionary is empty, return False
len()	Return the length (the number of items) in the dictionary
cmp()	Compares items of two dictionaries
sorted()	Return a new sorted list of keys in the dictionary

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Dictionary example



```
CS-16 - [~/PycharmProjects/CS-16] - .../dictionaries.py - PyCharm Community Edition 2016.3
File Edit View Navigate Code Refactor Run Tools VCS Window Help

CS-16 dictionaries.py dictionaries
file.py x break_cont.py x functions.py x data_str.py x dictionaries.py x

1 my_dictionary = {'name': 'John', 'age': 17}
2
3 # update value
4 my_dictionary['age'] = 16
5 print(my_dictionary)
6
7 # add item
8 my_dictionary['address'] = 'Oxford'
9 print(my_dictionary)
10
11 #Iterate through a dictionary
12
13 for key,value in my_dictionary.items():
14     print(key,value)

Run: nested_if dictionaries
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/dictionaries.py
{'age': 16, 'name': 'John'}
{'age': 16, 'name': 'John', 'address': 'Oxford'}
age 16
name John
address Oxford

Process finished with exit code 0

9:1 LF UTF-8
```

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Lesson 16: Lab 4 project

Learning objectives:

- To work on your project

Python project

In this lab session, you should work on your chosen project and prepare for your presentation.

Talk to your teacher if you have any concerns or questions.

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Lesson 17: Strings and REGEX

Learning objectives:

- To learn how strings work in Python and practice the different string methods
- To discover the limitless possibilities regular expressions have to offer with pattern matching

Lesson outline:

Strings

Your teacher will:

- Introduce strings
- Demonstrate some string methods, isupper, upper, lower, endswith, isdigit, isalpha, startswith...

Exercise 33

You have 20 minutes to:

- Write a Python program that checks whether a string is a palindrome or not. A palindrome is a string which is the same read forwards or backwards. For example, "dad" is the same in forward or reverse direction
- Remove punctuation from a string
- Write a program that sorts 5 words of your choice in alphabetical order

Regular expressions

Your teacher will:

- Define regular expressions and why they would be useful
- Explain the regex syntax for Python
- Demonstrate using regex to check for a particular pattern in a string

Exercise 34

You have 20 minutes to:

- Create a regex expression that checks if a car plate number given by the user follows the naming pattern followed in their country

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Strings and REGEX

Strings

String methods

Method	Description	Return type
s.count(s1)	Returns the number of non-overlapping occurrences of s1 in the string s	int
s.lower()	Returns a copy of the string with all letters converted to lowercase	str
s.upper()	Returns a copy of the string with all letters converted to uppercase	str
s.islower()	Returns True if all characters in the string are lowercase	bool
s.isupper()	Returns True if all characters in the string are uppercase	Bool
s.find(s1)	Returns the index of the first occurrence of s1 in the string s, or -1 if char does not occur (case-sensitive)	int
s.find(s1, b)	Returns the index of the first occurrence of s1 at or after index b in the string, or -1	int
s.find(s1, b, e)	Returns the index of the first occurrence of s1 between b (inclusive) and e(exclusive) in the string, or -1	int
s.index()	Returns the index of s1 in the string s, it gives an error if s1 does not occur	int
s.replace(o, n)	Returns a copy of the string with all occurrences of substring o replaced with string n	str
s.startswith(s1)	Returns True if s starts with s1	bool
s.endswith(s1)	Returns True if s ends with s1	bool
s.lstrip()	Returns a copy of the string with leading whitespace removed	str
s.rstrip()	Returns a copy of the string with trailing whitespace removed	str
s.strip()	Returns a copy of the string with leading and trailing whitespace removed	str
s.isdigit()	Returns True if all characters of s are digits	bool
s.isalpha()	Returns True if all characters of s are alphabetic	bool
s.isalnum()	Returns True if all characters of s are alphanumeric	bool

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Strings example

The screenshot shows the PyCharm Community Edition 2016.3 interface. The main editor window displays the file `strings.py` with the following Python code:

```
1 str1 = 'Hello'
2 str2 = 'World!'
3
4 # using +
5 print('str1 + str2 = ', str1 + str2)
6 # using *
7 print('str1 * 3 =', str1 * 3)
8 print(str1.upper())
9 print(str1.capitalize())
10 print(str1.lower())
11
12
13
```

The left sidebar shows a project view with files like `data_str.py`, `dictionaries.py`, `elif.py`, `for_loop.py`, `functions.py`, `hello_pytho`, `Identity_me`, `if.py`, `if_else.py`, `logic.py`, `nested_if.p`, `operators.p`, `strings.py`, and `while.py`. The `strings.py` file is selected.

At the bottom, the Run window shows the execution of `strings.py` using `/usr/bin/python3.5`. The output is as follows:

```
str1 + str2 = HelloWorld!
str1 * 3 = HelloHelloHello
HELLO
Hello
hello
```

The Run window also indicates "Process finished with exit code 0". The status bar at the bottom shows the time as 9:1, the file encoding as n/a, and the character set as UTF-8.

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Source: <https://xkcd.com/208/>

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Regular expression symbols in Python

<code>^</code>	Matches the beginning of a line
<code>\$</code>	Matches the end of the line
<code>.</code>	Matches any character
<code>\s</code>	Matches whitespace
<code>\S</code>	Matches any non-whitespace character
<code>*</code>	Repeats a character zero or more times
<code>*?</code>	Repeats a character zero or more times (non-greedy)
<code>+</code>	Repeats a character one or more times
<code>+?</code>	Repeats a character one or more times (non-greedy)
<code>[aeiou]</code>	Matches a single character in the listed set
<code>[^XYZ]</code>	Matches a single character not in the listed set
<code>[a-z0-9]</code>	The set of characters can include a range
<code>(</code>	Indicates where string extraction is to start
<code>)</code>	Indicates where string extraction is to end

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Lesson 18: File operations

Learning objectives:

- To learn how to work with files in Python
- To be comfortable with creating, opening and closing files
- To understand the difference between the file opening modes – read and write

Lesson outline:

Files

Your teacher will:

- Define files
- Demonstrate how to open and close a file in Python
- Explain the different file opening modes – read, write, append, exclusive, text, binary, update

Exercise 35

You have 20 minutes to:

- Create a text file in any text editor and save it in the same location as the Python file
- Open and close the file in Python with either the read or append mode – write will empty its contents

Reading files

Your teacher will demonstrate the different ways in which you can read information from a file in Python.

Exercise 36

You have 20 minutes to:

- Write a program that uses the text file created in Exercise 35, read the content of the file and print them out using Python

Writing to file

Exercise 37

You have 20 minutes to:

- Practise write mode, use the same text file created in Exercise 35 and 36 and add more information to the file by writing to it using Python
- Open a file in write mode and add information to it

More file methods

Your teacher will explain more file methods (such as `close()` and `tell()` methods) available in Python.

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Extra

Using Python, create a text file to store user names and passwords. Ask the user to enter their user name and password. The system should read in the details from the file and then return one of these messages:

- Your login is successful (correct username and password)
- Incorrect password (username exists but password does not match)
- This username does not exist (username does not exist in the file)

Additional supporting materials

File operations

Python file modes

Mode	Description
'r'	Open a file for reading (default)
'w'	Open a file for writing. Creates a new file if it does not exist or truncates the file if it exists
'x'	Open a file for exclusive creation. If the file already exists, the operation fails
'a'	Open for appending at the end of the file without truncating it. Creates a new file if it does not exist
't'	Open a file in text mode (default)
'b'	Open a file in binary mode
'+'	Open a file for updating (reading and writing)

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Python file methods

Method	Description
<code>close()</code>	Close an open file. It has no effect if the file is already closed
<code>detach()</code>	Separate the underlying binary buffer from the TextIOBase and return it
<code>fileno()</code>	Returns an integer number (file descriptor) of the file
<code>flush()</code>	Flush the write buffer of the file stream
<code>isatty()</code>	Returns True if the file stream is interactive
<code>read(n)</code>	Read at most n characters from the file. Reads till end of file if it is negative or None
<code>readable()</code>	Returns True if the file stream can be read from
<code>readline(n=-1)</code>	Read and return one line from the file. Reads in at most n bytes if specified
<code>readlines(n=-1)</code>	Read and return a list of lines from the file. Reads in at most n bytes/characters if specified
<code>seek(offset,from=SEEK_SET)</code>	Change the file position to offset bytes, in reference to from(start, current, end)
<code>seekable()</code>	Returns True if the file stream supports random access
<code>tell()</code>	Returns the current file location
<code>truncate(size=None)</code>	Resize the file stream to size bytes. If size is not specified, resize to current location
<code>writable()</code>	Returns True if the file stream can be written to
<code>write(s)</code>	Write string s to the file and return the number of characters written
<code>writelines(lines)</code>	Write a list of lines to the file

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Lessons 19 and 20: Presentations

The presentation is about your project which should last between 10 and 12 minutes. You may use PowerPoint if you wish. Your aim is to teach your peers something about your Python software development process and the functionality of your program. Some things you should consider include:

- What problem were you trying to solve?
- Did you use a clever method to simplify the problem?
- Did you use a particularly efficient algorithm?
- How would you adjust the process for a more realistic problem?
- What might you have done differently if you had had more time/computing power?

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Lesson 21: Lab 5 project

Learning objectives:

- To complete your project

Your Python project and written report are due in today!

In this lab lesson, you are expected to complete your Python project and express any concerns you might have.

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Lesson 22: Debugging

Learning objectives:

- To understand the need for a debugging system
- To be able to analyse real-world examples of bugs and software errors
- To appreciate the variety of possible problems and the one last bug syndrome
- To be able to apply a variety of debugging techniques to your own code
- To be able to decide which techniques are most appropriate in a given context
- To learn how to handle exceptions in your code
- To utilise good layout and programming practice in your code to facilitate debugging

Lesson outline:

Debugging

Your teacher will:

- Introduce debugging and its importance in the production of clean code
- Discuss the different types of errors
- Demonstrate different debugging methods including how to use breakpoints

Exercise 38

You have 25 minutes to:

- List and discuss some common code errors made by programmers
- Find out where the words “bug” and “debugging” in programming originate from
- Discuss why we use bug-tracking tools in programming and list 3 types of bug tracking tools/debuggers
- Try debugging the program (in debugging1.py) below and run the bug free codes

Exception handling

Your teacher will:

- Explain the need for exceptions and some Python built-in exceptions
- Demonstrate how you can define your own user-defined exceptions
- Demonstrate how to create a try-catch block

Exercise 39

You have 25 minutes to:

- Identify possible cases in your previous programs where user errors may occur
- Include the “try and except” blocks to catch exceptions in 2 of your programs from

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your previous exercises to work with. File operations, user inputs and numerical operations are just examples of catchable exceptions

Extra

Write a python program that calculates students' grades. The code overleaf (in debugging2.py) has errors. The correct version will average 3 numeric exam grades, return an average test score, a corresponding letter grade and a message stating whether the student is passing.

Average	Grade
90+	A
80-89	B
70-79	C
60-69	D
0-59	F

Exams: 89, 96, 90

Average: 91.6666667

Grade: A

Student is passing.

Exams: 50, 51, 0

Average: 33

Grade: F

Student is failing.

Additional supporting materials

Debugging

The code below has errors.

```
CS-16 - [~/PycharmProjects/CS-16] - .../debugging1.py - PyCharm Community Edition 2016.3
File Edit View Navigate Code Refactor Run Tools VCS Window Help
CS-16 > debugging1.py > debugging1
tutions.py x data_str.py x dictionaries.py x strings.py x debugging1.py x
CS-16 ~/Pychar
1 authors = {
2     "JK Rowling": "1870",
3     "JJ Abrahms": "1863",
4     "GR Martins": "1882",
5     "George Lucas": "1889"
6
7     for author date in authors.items():
8         print(author, date)
9 }
10
Run: nested_if debugging1
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/debugging1.py
File "/home/fredah/PycharmProjects/CS-16/debugging1.py", line 7
    for author date in authors.items():
        ^
SyntaxError: invalid syntax
Process finished with exit code 1
```

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Fix the errors and run the code. The correct code should output as shown below

The Run window shows the output of the program debugging1.py. The output is:

```
GR Martins 1882
JJ Abrahms 1863
George Lucas 1889
JK Rowling 1870
```

Below the output, it states: "Process finished with exit code 0".

The PyCharm IDE shows the file debugging2.py. The code is as follows:

```
22 exam_one = int(input("Input exam grade one: "))
23 exam_two = input("Input exam grade two: ")
24 exam_three = str(input("Input exam grade three: "))
25 grades = [exam_one exam_two exam_three]
26 sum = 0
27 for grade in grade:
28     sum = sum + grade
29 avg = sum / len(grades)
30 if avg >= 90:
31     letter_grade = "A"
32 elif avg >= 80 and avg < 90:
33     letter_grade = "B"
34 elif avg > 69 and avg < 80:
35     letter_grade = "C"
36 elif avg <= 69 and avg >= 65:
37     letter_grade = "D"
38 elif:
39     letter_grade = "F"
40
41 for grade in grades:
42     print("Exam: " + str(grade))
43     print("Average: " + str(avg))
44     print("Grade: " + letter_grade)
45 if letter_grade is "F":
46     print("Student is failing.")
47 else:
48     print("Student is passing.")
```

A tooltip indicates a syntax error: "Colon expected" at line 34. The Run window shows the following error:

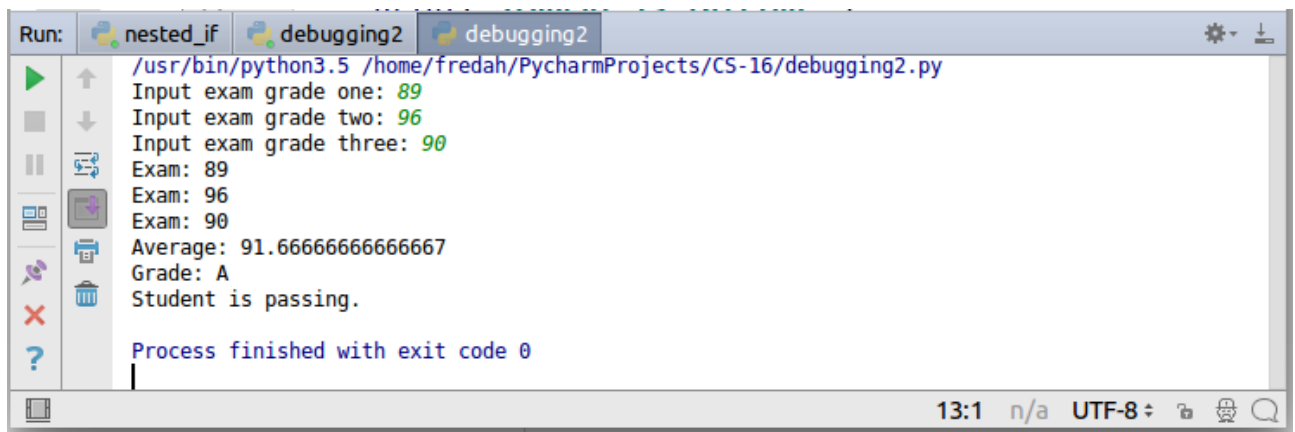
```
File "/home/fredah/PycharmProjects/CS-16/debugging2.py", line 23
exam_two = input("Input exam grade two: ")
^
SyntaxError: invalid syntax
```

The Run window also shows: "Process finished with exit code 1".

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Correct output for input 89, 96 and 90 is shown below.



```
Run: nested_if debugging2 debugging2
/usr/bin/python3.5 /home/fredah/PycharmProjects/CS-16/debugging2.py
Input exam grade one: 89
Input exam grade two: 96
Input exam grade three: 90
Exam: 89
Exam: 96
Exam: 90
Average: 91.66666666666667
Grade: A
Student is passing.

Process finished with exit code 0
```

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Exception handling

Python built-in exceptions

Exception	Cause of error
AssertionError	Raised when assert statement fails
AttributeError	Raised when attribute assignment or reference fails
EOFError	Raised when the input() functions hits end-of-file condition
FloatingPointError	Raised when a floating point operation fails
GeneratorExit	Raise when a generator's close() method is called
ImportError	Raised when the imported module is not found.
IndexError	Raised when index of a sequence is out of range
KeyError	Raised when a key is not found in a dictionary
KeyboardInterrupt	Raised when the user hits interrupt key (Ctrl+c or delete)
MemoryError	Raised when an operation runs out of memory
NameError	Raised when a variable is not found in local or global scope
NotImplementedError	Raised by abstract methods
OSError	Raised when system operation causes system related error
OverflowError	Raised when result of an arithmetic operation is too large to be represented
ReferenceError	Raised when a weak reference proxy is used to access a garbage collected referent
RuntimeError	Raised when an error does not fall under any other category
StopIteration	Raised by next() function to indicate that there is no further item to be returned by iterator
SyntaxError	Raised by parser when syntax error is encountered
IndentationError	Raised when there is incorrect indentation

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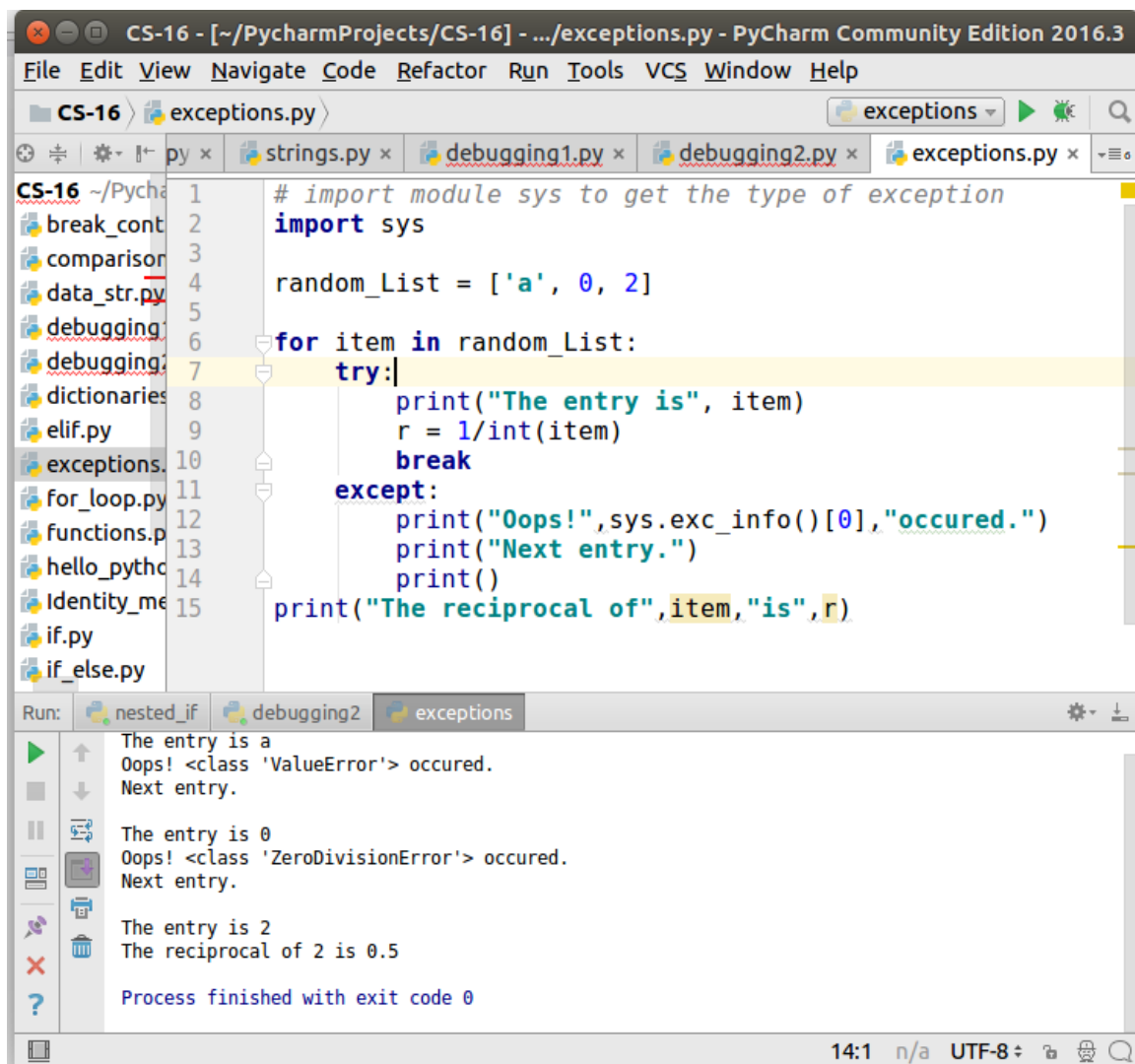
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TabError	Raised when indentation consists of inconsistent tabs and spaces
SystemError	Raised when interpreter detects internal error
SystemExit	Raised by sys.exit() function
TypeError	Raised when a function or operation is applied to an object of incorrect type
UnboundLocalError	Raised when a reference is made to a local variable in a function or method, but no value has been bound to that variable
UnicodeError	Raised when a Unicode-related encoding or decoding error occurs
UnicodeEncodeError	Raised when a Unicode-related error occurs during encoding
UnicodeDecodeError	Raised when a Unicode-related error occurs during decoding
UnicodeTranslateError	Raised when a Unicode-related error occurs during translating
ValueError	Raised when a function gets argument of correct type but improper value
ZeroDivisionError	Raised when second operand of division or modulo operation is zero

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Exception handling example



The screenshot shows the PyCharm IDE with a project named 'CS-16'. The main editor window displays the file 'exceptions.py' with the following Python code:

```
1 # import module sys to get the type of exception
2 import sys
3
4 random_List = ['a', 0, 2]
5
6 for item in random_List:
7     try:
8         print("The entry is", item)
9         r = 1/int(item)
10        break
11    except:
12        print("Oops!", sys.exc_info()[0], "occured.")
13        print("Next entry.")
14        print()
15 print("The reciprocal of", item, "is", r)
```

The 'Run' tab at the bottom shows the execution output:

```
The entry is a
Oops! <class 'ValueError'> occured.
Next entry.

The entry is 0
Oops! <class 'ZeroDivisionError'> occured.
Next entry.

The entry is 2
The reciprocal of 2 is 0.5

Process finished with exit code 0
```

The status bar at the bottom indicates the time is 14:1, the encoding is n/a, and the character set is UTF-8.

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Lesson 23: Software testing

Learning objectives:

- To understand the need for software testing
- To know the difference between white, black and grey box testing

Lesson outline:

Poorly-developed Python program

Your teacher will:

- Demonstrate a poorly-developed Python program
- Introduce the concept behind software testing and its importance
- Highlight the different things to look out for when testing your software

Types of testing

Your teacher will briefly describe the difference between white, black and grey box testing cycles and how that results in software evolution.

Exercise 40

You have 10 minutes to:

- Read through the article on software testing:
http://www.tutorialspoint.com/software_testing/software_testing_methods.htm
- Provide three differences between white, grey and black box testing
- Look through an app on your phone or a website of your choice to find defects in the design or functionalities. Discuss defects in pairs

Test cases

Your teacher will:

- Define a test case and its use
- Demonstrate how to build a test case for your software product
- Show how these tests can be carried out and how changes are made to the programs to make it run better
- Demonstrate the perfectly working program after testing and rectifying the errors

Exercise 41 (group task)

You have 10 minutes to:

- Develop 3 real-life test cases/scenarios for the Apple Music or Google Play Music App.
Write a brief description of each of these scenarios

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- Write a defect report or a software testing report, indicating areas that have been tested (such as the GUI design, the app performance, functionality and usability)
- Provide suggestions to improve the user's experience or quality of the app
- Present your answers to the class
- Discuss other effective methods of testing

Additional supporting materials

To ensure quality of your finished product, it is essential that you, as a developer, perform a series of tests on your software. There are different kinds of software testing methods, namely black box, white box and grey box. These tests are not mutually exclusive and different types of test can be carried out on the same piece of software.

Please find time to go through this article on the different types:

http://www.tutorialspoint.com/software_testing/software_testing_methods.htm

Name three differences between black, white, and grey box testing.

1. _____

2. _____

3. _____

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Lesson 24: Evolution and maintenance

Learning objectives:

- To understand the full software development lifecycle
- To be able to use software evolution and maintenance to understand why the cycle never ends

Lesson outline:

Software development lifecycle

Your teacher will:

- Introduce the software development lifecycle
- Discuss the different models of software development, giving pros and cons for each

Exercise 42

- In groups, you have 20 minutes to defend the software development model assigned to you. Talk about its pros and cons. Is it the best development method?
- What does security mean in software development, and at what phase should developers start implementing security in their software products?

Software evolution and maintenance

Your teacher will:

- Explain in detail the maintenance phase in the lifecycle and how that results in software evolution
- Elaborate why it is important for developers to have a well-structured maintenance method to allow for feedback from users

Exercise 43

You have 10 minutes to:

- Discuss the various types of software maintenance
- Give examples of great software maintenance cases and explain why you support it
- Give examples of terrible software maintenance cases in real life
- What does software re-engineering/reverse engineering mean? What are some examples of tools used?

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Open question and answer session

- Ask any Python-related questions you have
- Your peers have the chance to answer your questions, and if they are unable to, your teacher will answer

Lessons 25 and 26: Lab – Python challenge

Learning objectives:

- To have fun with Python challenges

Lesson outline:

Python challenge

You will be put in pairs or groups of 3 and tasked with completing some Python challenges. Some are harder than others, so do not worry if you struggle. These last two lessons are designed to test your problem-solving skills and coding abilities in Python that you have developed over the last two weeks.

Materials for Python challenge

Finding HCF and LCM

The highest common factor (HCF) or greatest common divisor (GCD) of two numbers is the largest positive integer that perfectly divides the two given numbers. For example, the HCF of 12 and 14 is 2.

The lowest common multiple (LCM) of two or more numbers is the smallest positive integer that perfectly divides the given numbers. For example, the LCM of 3 and 5 is 15 and the LCM of 7, 4 and 3 is 84.

Breaking down an amount into the fewest possible notes and coins

The British pound has the following denominations in notes and coins:

Pounds = [50, 20, 10, 5, 2, 1, 0.5, 0.25, 0.1, 0.05, 0.02, 0.01, 0.005]

Given the amount 47.56, the smallest breakdown using the British pound notes and coins consists of

{20: 2, 5:1, 2:1, 0.5, 0.1:1}

Ancient Chinese puzzle

Write a program to solve a classic ancient Chinese puzzle. You count 35 heads and 94 legs among the chickens and rabbits in a farm. How many rabbits and how many chickens do you have?

Remove duplicates from the list

With the given list [12,24,35,24,88,120,155,88,120,155], write a program to print this list after removing all duplicate values with the original order reserved.

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Three summation problem

Given a list [2, 4, 8, 10, -7, -10, -25, -1, 0, 1, 2, -1, -4], find three unique elements which, when added together, add up to 0.

For the given list, the following elements add up to 0:

[[-10, 0, 10], [-10, 2, 8], [-7, -1, 8], [-4, 0, 4], [-4, 2, 2], [-1, -1, 2], [-1, 0, 1]]

Maze navigation

In this challenge, you are required to figure out if the maze is solvable.

Assume a maze is given by a list of lists:

```
maze1 = [[0, 1, 1, 1, 0, 0, 0, 1],  
          [0, 1, 0, 1, 0, 1, 0, 1],  
          [0, 0, 0, 0, 0, 0, 0, 0],  
          [1, 1, 1, 1, 0, 0, 1, 0],  
          [1, 0, 0, 0, 1, 0, 1, 0],  
          [0, 0, 1, 0, 1, 0, 1, 0],  
          [0, 1, 1, 0, 0, 0, 0, 0],  
          [0, 0, 0, 1, 1, 0, 1, 0]]
```

The cells that contain 0 are passable and the cells with 1 are walls. A person starting at maze[0][0] (top left) can move up, down, left or right, but only to other passable cells. The question is whether

the maze can be solved. Assume the goal is to get to the bottom right corner (e.g., maze[7][7]. maze1 given above is solvable, since there is a path from the top left to the bottom right).

As another example, maze2 below is not solvable (there is no path from the top left to the bottom right):

```
maze2 = [[0, 1, 0, 0, 0, 0, 0, 1],  
          [0, 0, 1, 1, 0, 1, 0, 1],  
          [1, 1, 0, 1, 0, 1, 0, 0],  
          [1, 1, 0, 1, 0, 0, 0, 1],  
          [1, 0, 0, 1, 0, 1, 1, 1],  
          [0, 0, 1, 0, 0, 1, 1, 1],  
          [0, 1, 1, 0, 0, 1, 0, 1],  
          [0, 0, 0, 0, 1, 1, 0, 0]]
```

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Create a function that takes arguments; a 2D maze list, a starting point and an end point (goal). The function should return True if the maze is solvable and False if not for the given starting and ending position.

The player can only move up, down, left and right; they cannot pass through walls and cannot move past the edges of the maze.

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Ways to change money

Consider an imaginary currency Σ which has 7 different denominations in circulation: the $\Sigma 1$ coin and the $\Sigma 5$, $\Sigma 10$, $\Sigma 20$, $\Sigma 50$, $\Sigma 100$ and $\Sigma 200$ notes. We ignore the denominations less than $\Sigma 1$

Implement the following function such that, when given an amount of money, it returns the number of ways one could make change for that amount of money. For example, there are exactly six ways to make change for $\Sigma 15$:

- $15 \times \Sigma 1$
- $1 \times \Sigma 5 + 10 \times \Sigma 1$
- $2 \times \Sigma 5 + 5 \times \Sigma 1$
- $3 \times \Sigma 5$
- $\Sigma 10 + 5 \times \Sigma 1$
- $\Sigma 10 + \Sigma 5$

so should return 6.

Again, since the only possible inputs are positive integers, your function should raise `ValueErrors` for negative amounts and `TypeError`s for non-integer inputs.

Mix strings

You need to create a method which will mix two strings of length m and n respectively. The function will concatenate n versions of the first string, each time substituting successive characters of it with successive characters of the second string. For example:

`mix("12345", "HELLO")` $\rightarrow$ "H23451E34512L45123L51234O"

`mix("abcd", "wxyz")` $\rightarrow$ "wbc daxcdabydabcz"

The returned string will always be of $m \times n$ length. You may assume that the first string will always be of the same length or longer than the second ($m \geq n$).

Substrings

Write a function called `substrings` that will return a set of all of the substrings of a given string s , including s itself and the empty string. For example:

`substrings('abc')` = {'', 'c', 'abc', 'a', 'ab', 'bc', 'b'}

`substrings('bobo')` $\Rightarrow$ {'bobo', '', 'bo', 'obo', 'o', 'ob', 'b', 'bob'}

`substrings('aaa')` = {'', 'a', 'aa', 'aaa'}

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